

Multilingual MT Dataset

Exploratory Data Analysis Report

Ar-En · Ar-Hi · Ar-Ur

Date: 08 June 2026

Phase 1 — Training Analysis: eda_train

Phase 2 — Cross-Split Analysis: eda_split

Sections covered:

Part 1 — Training Set Analysis (alignment, per-file stats, language aggregation)

Part 2 — Cross-Split Analysis (OOV, leakage, n-gram overlap, semantic similarity)

Part 3 — Overall Assessment

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Part 1: Training Set Analysis

1.1 Alignment Check

Pair	Source Lines	Target Lines	Delta	Status
Ar-En	21000	21000	0	PASS ✓
Ar-Hi	20100	20100	0	PASS ✓
Ar-Ur	20300	20300	0	PASS ✓

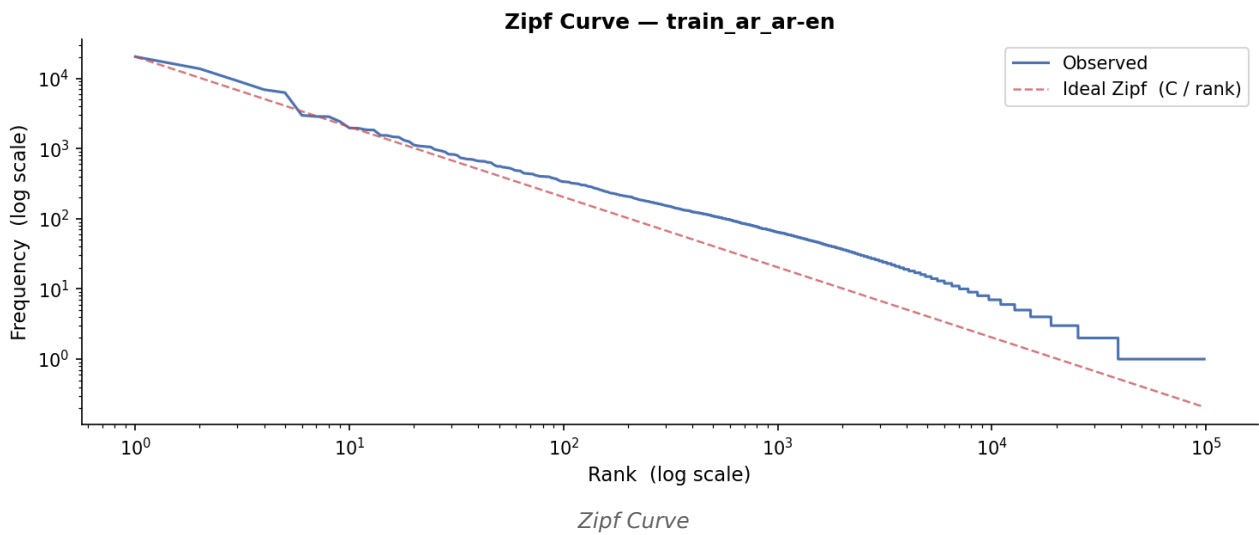
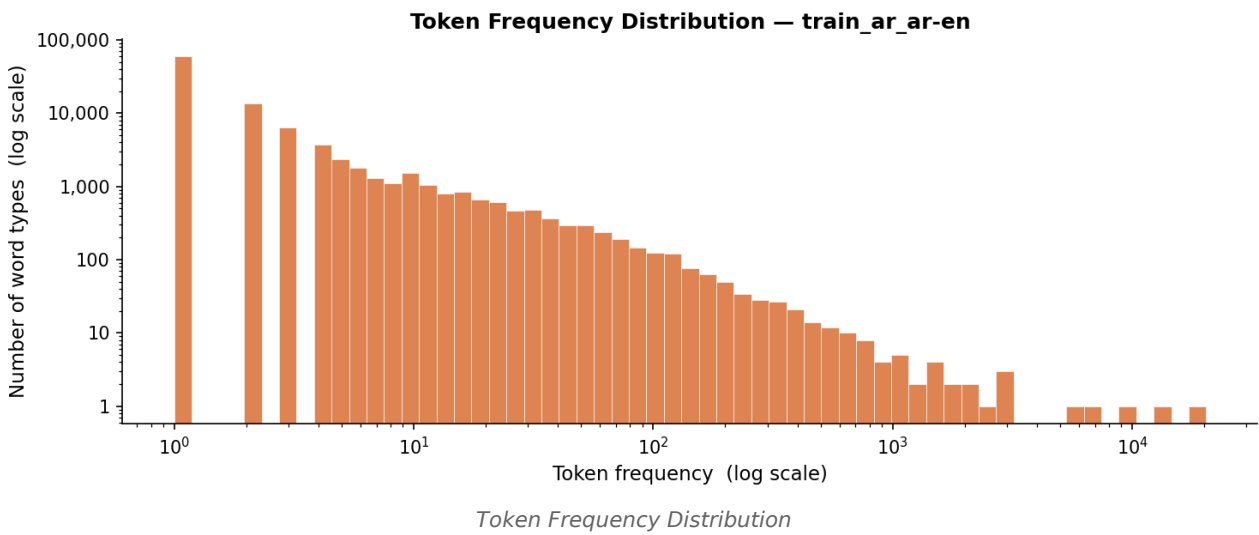
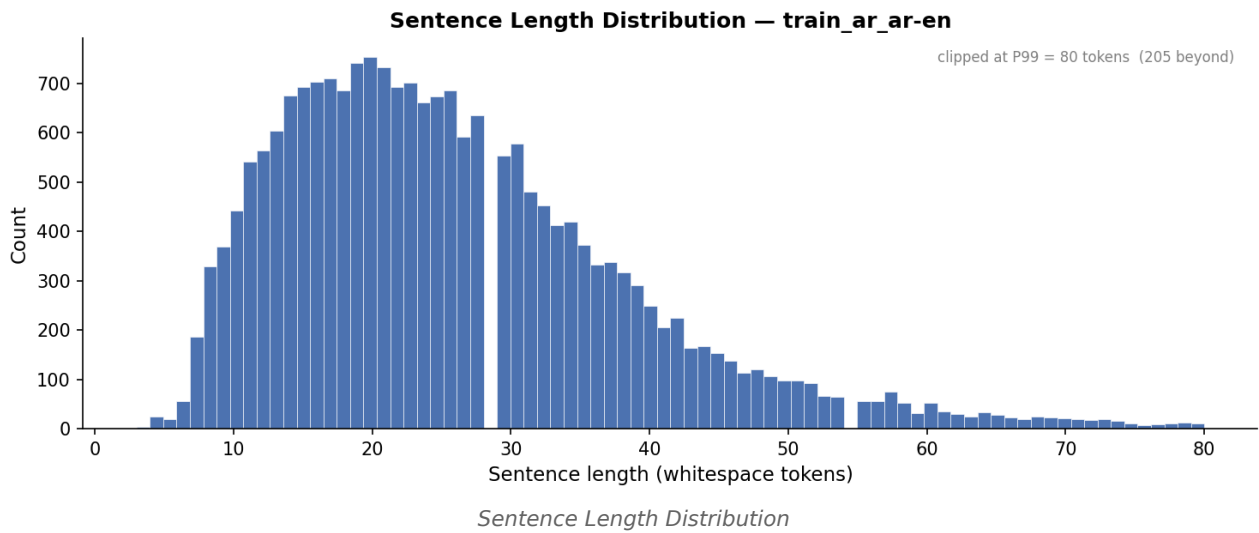
1.2 Per-File Summary

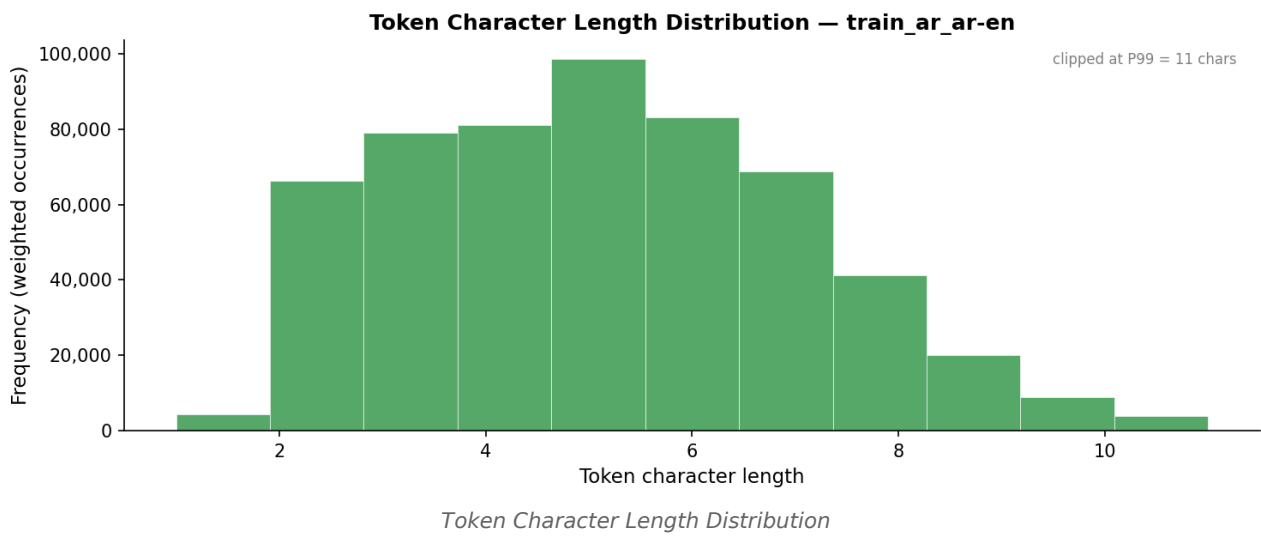
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train_210_210	210	210	0	0	0.0	26.1	24.1	3	120	14.4	52.1	80.1	558	980	0.1	592	135	124	563	5.1	558	980	0.1	592	
train_201_201	201	201	0	0	0.02	31.9	28.03	509	19.06	67.3	101.6	105.4	1308	582	0.1	580	133	121	544	5.1	539	959	0.1	580	
train_203_203	203	203	0	0	0.04	38.3	33.03	421	22.6	91.3	152.7	704	953	1704	1598	0.1	582	133	122	547	5.1	543	963	0.1	582
train_200_200	200	200	0	0	0.02	31.9	28.03	509	19.06	67.3	101.6	105.4	1308	582	0.1	580	133	121	544	5.1	539	959	0.1	580	
train_202_202	202	202	0	0	0.02	31.9	28.03	509	19.06	67.3	101.6	105.4	1308	582	0.1	580	133	121	544	5.1	539	959	0.1	580	

1.3 Per-File Detailed Statistics

File: train_ar_ar-en

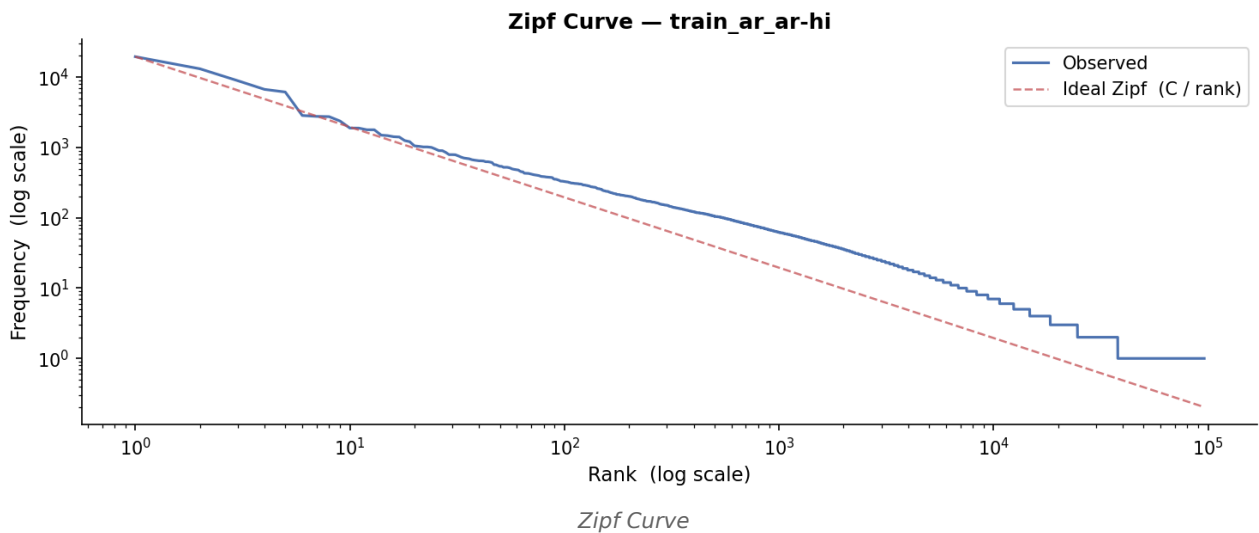
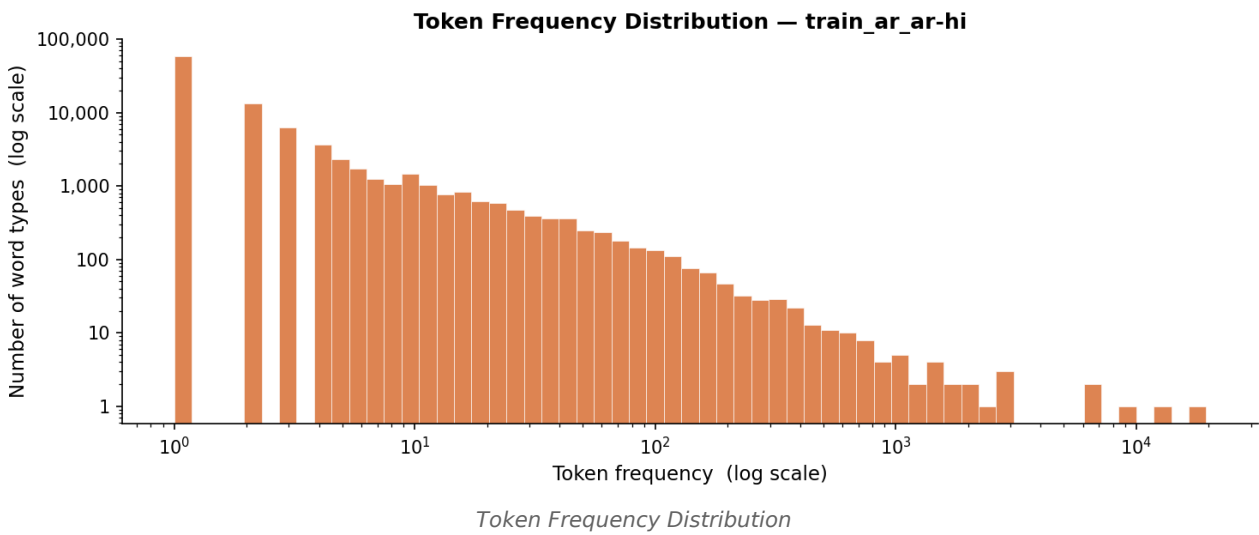
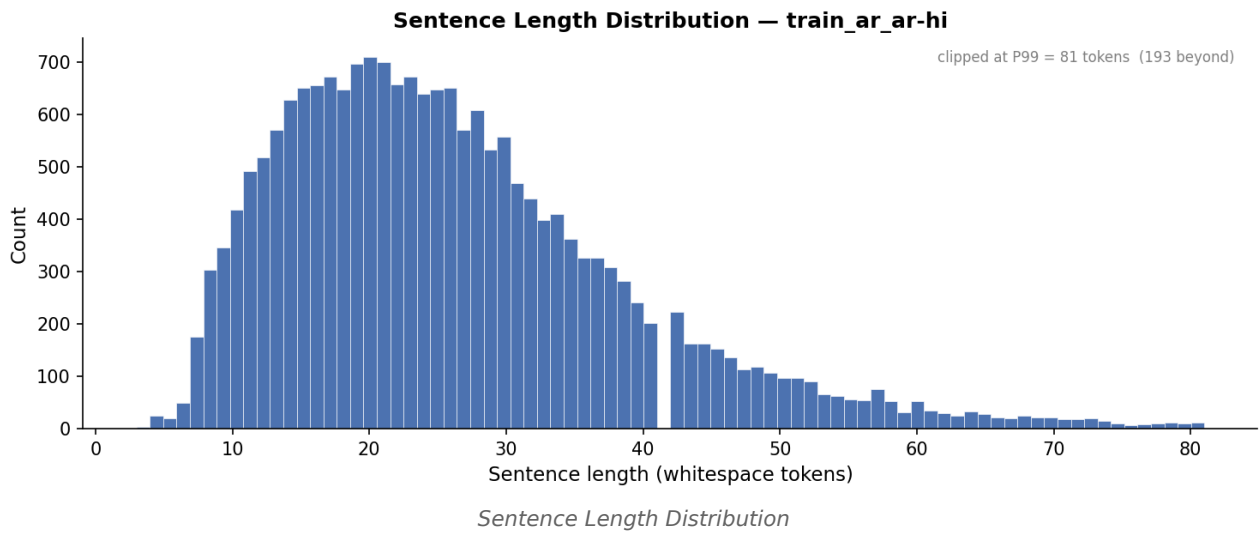
Total Sentences	21000
Unique Sentences	21000
Duplicate Sentences	0
Duplicate Pct	0.0000
Empty Sentences	0
Mean Len	26.5778
Median Len	24.0000
Min Len	3
Max Len	120
Std Len	14.4019
P95 Len	52.0000
P99 Len	80.0000
— Raw Vocab —	
Total Tokens	558133
Unique Tokens	98062
Ttr	0.1757
Freq Exactly 1	59252
Freq Exactly 2	13582
Freq 3 To 5	12461
Freq Over 100	563
Mean Token Char Len	5.1036
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125
— Norm Vocab —	
Total Tokens	558133
Unique Tokens	98062
Ttr	0.1757
Freq Exactly 1	59252
Freq Exactly 2	13582
Freq 3 To 5	12461
Freq Over 100	563
Mean Token Char Len	5.1036
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125

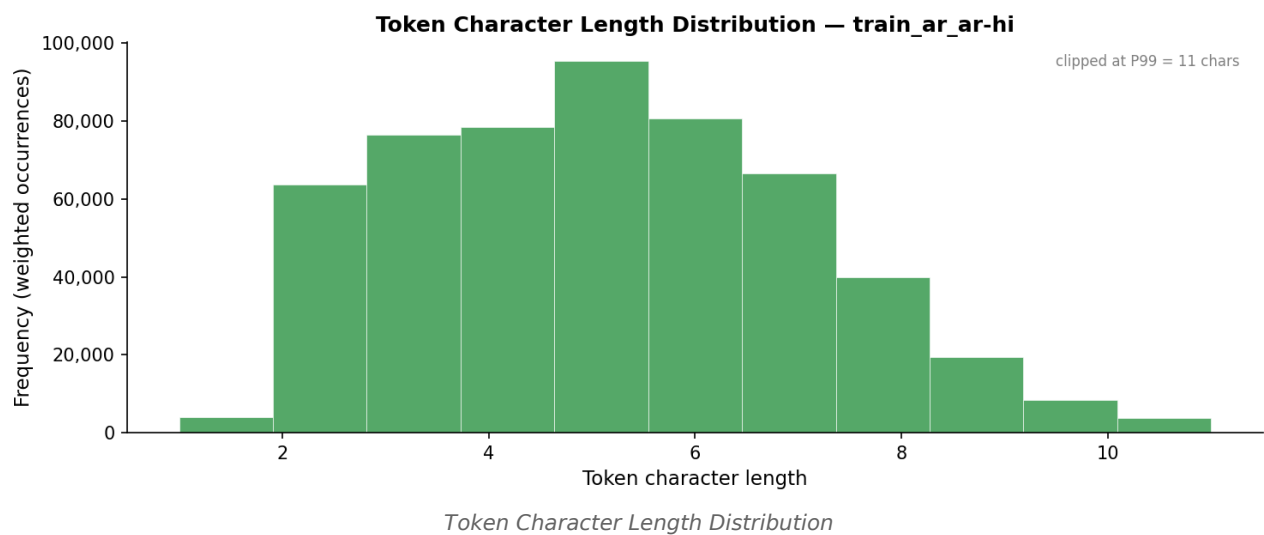




File: train_ar_ar-hi

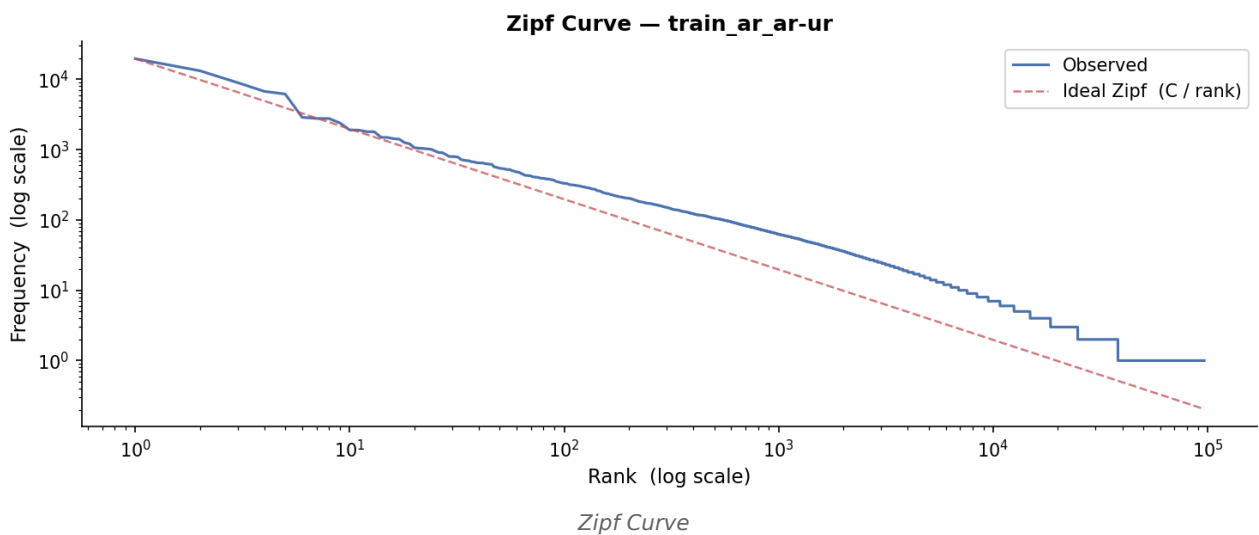
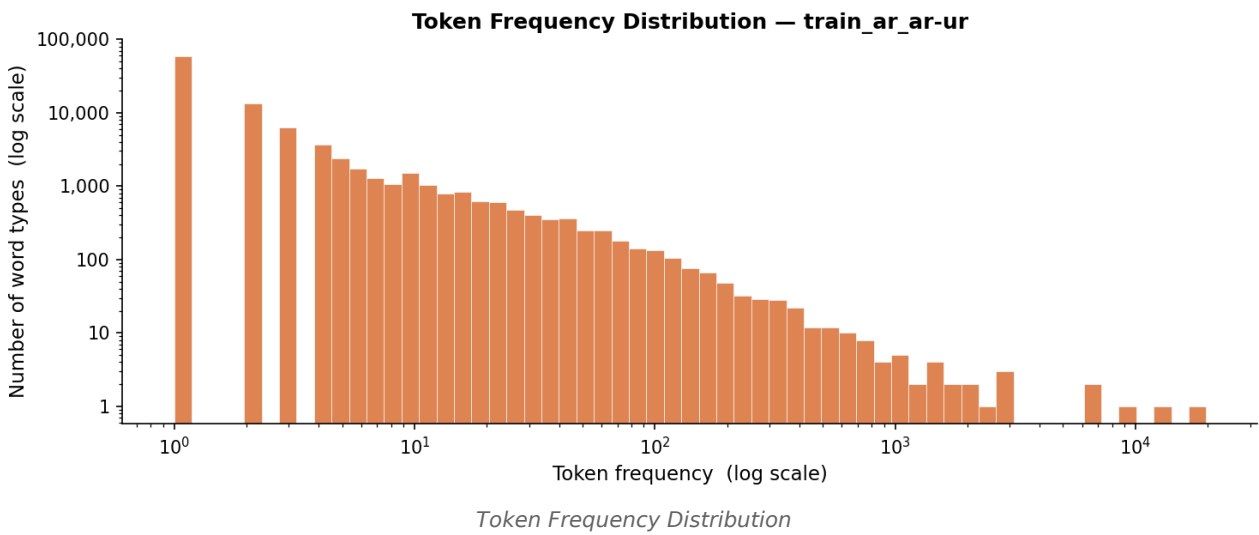
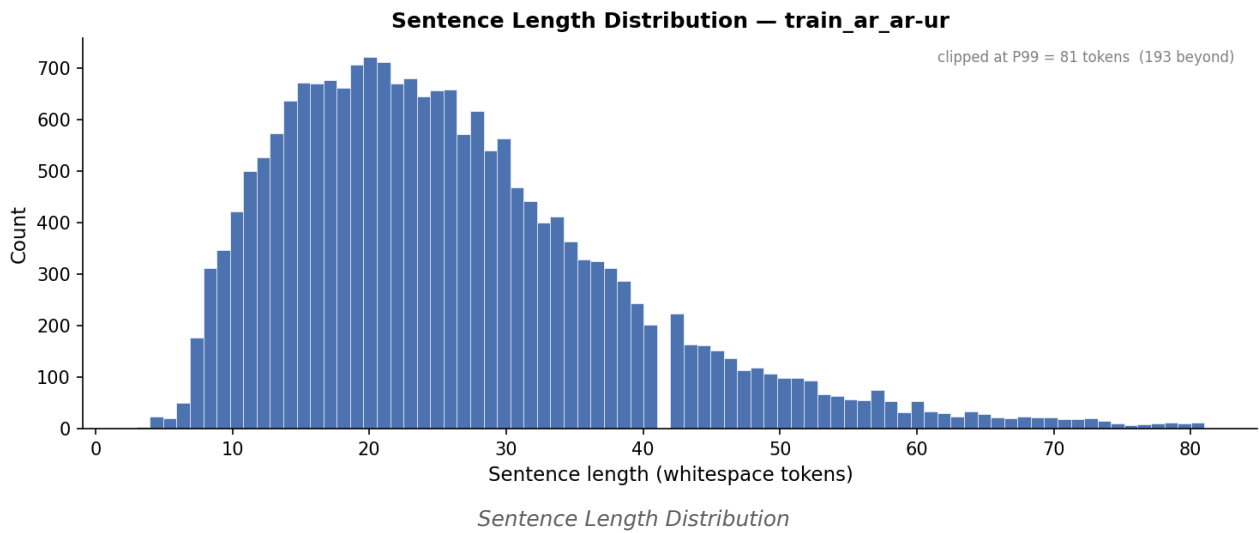
Total Sentences	20100
Unique Sentences	20100
Duplicate Sentences	0
Duplicate Pct	0.0000
Empty Sentences	0
Mean Len	26.8419
Median Len	24.0000
Min Len	3
Max Len	120
Std Len	14.5310
P95 Len	53.0000
P99 Len	81.0000
— Raw Vocab —	
Total Tokens	539522
Unique Tokens	95932
Ttr	0.1778
Freq Exactly 1	58038
Freq Exactly 2	13304
Freq 3 To 5	12190
Freq Over 100	544
Mean Token Char Len	5.1062
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125
— Norm Vocab —	
Total Tokens	539522
Unique Tokens	95932
Ttr	0.1778
Freq Exactly 1	58038
Freq Exactly 2	13304
Freq 3 To 5	12190
Freq Over 100	544
Mean Token Char Len	5.1062
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125

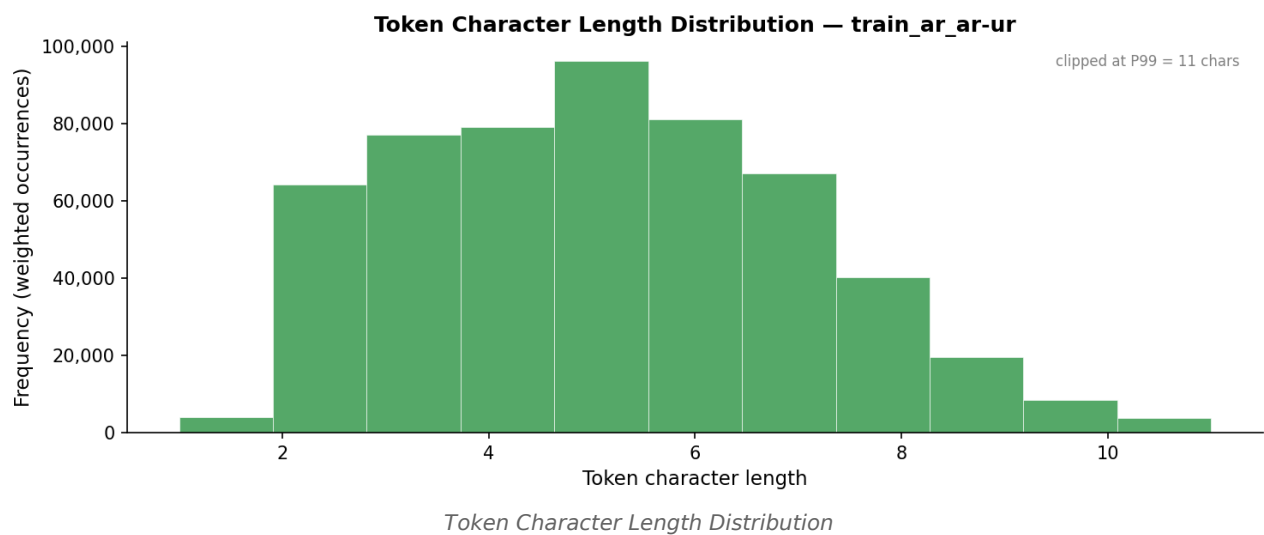




File: train_ar_ar-ur

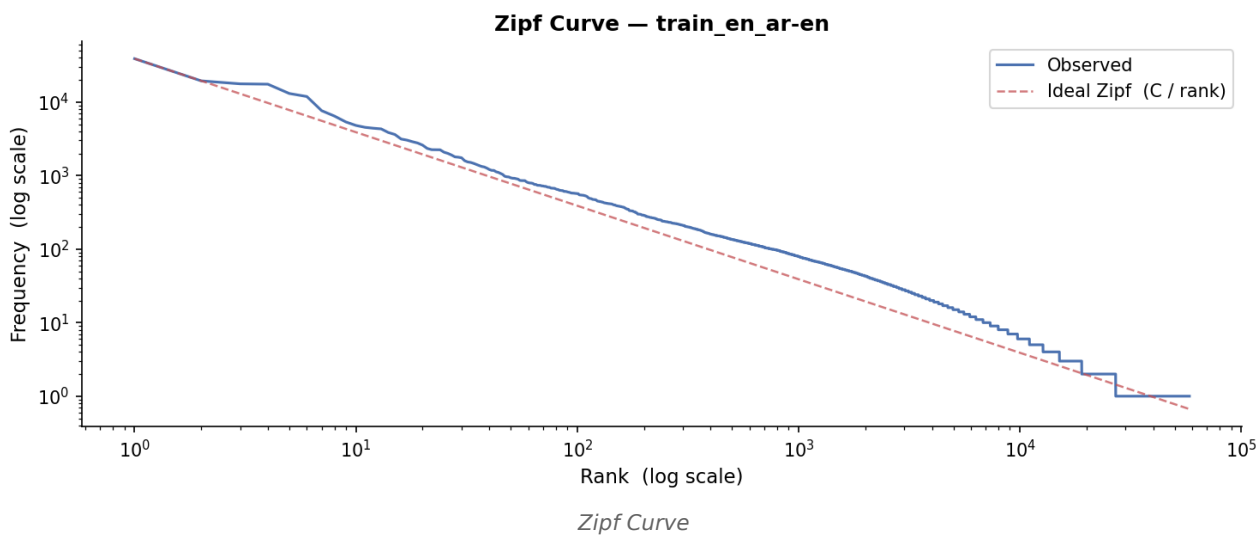
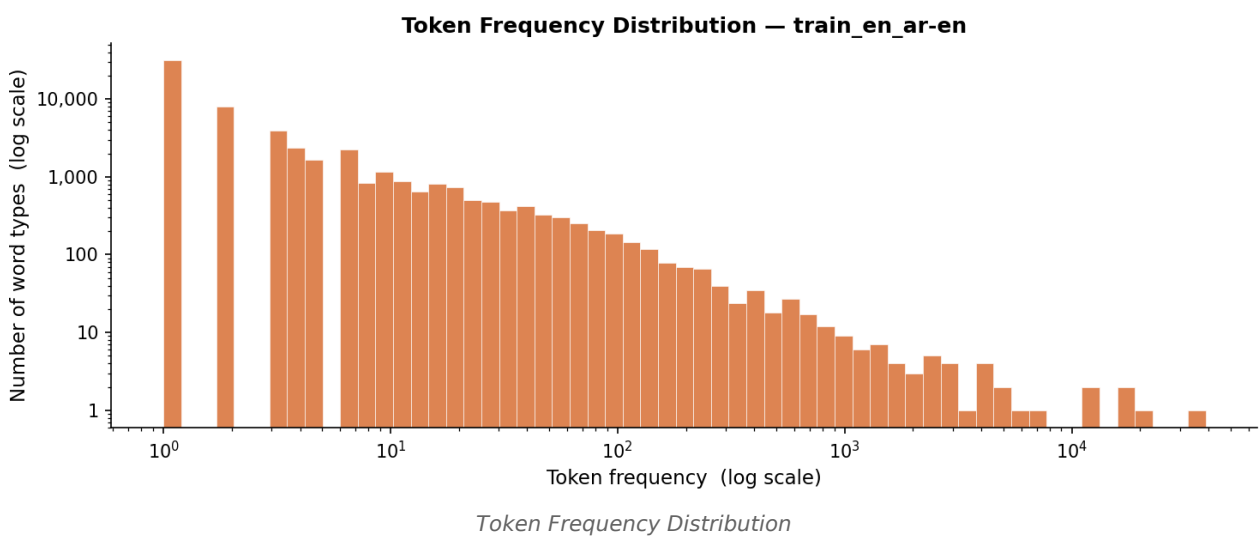
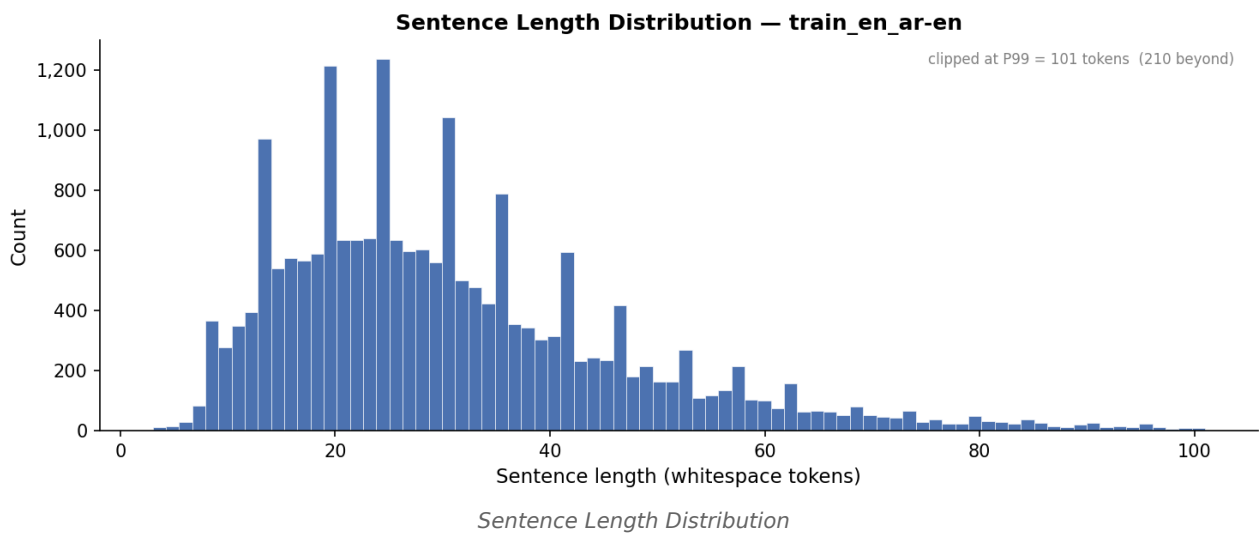
Total Sentences	20300
Unique Sentences	20300
Duplicate Sentences	0
Duplicate Pct	0.0000
Empty Sentences	0
Mean Len	26.7918
Median Len	24.0000
Min Len	3
Max Len	120
Std Len	14.5036
P95 Len	53.0000
P99 Len	81.0000
— Raw Vocab —	
Total Tokens	543873
Unique Tokens	96320
Ttr	0.1771
Freq Exactly 1	58213
Freq Exactly 2	13364
Freq 3 To 5	12256
Freq Over 100	547
Mean Token Char Len	5.1059
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125
— Norm Vocab —	
Total Tokens	543873
Unique Tokens	96320
Ttr	0.1771
Freq Exactly 1	58213
Freq Exactly 2	13364
Freq 3 To 5	12256
Freq Over 100	547
Mean Token Char Len	5.1059
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125

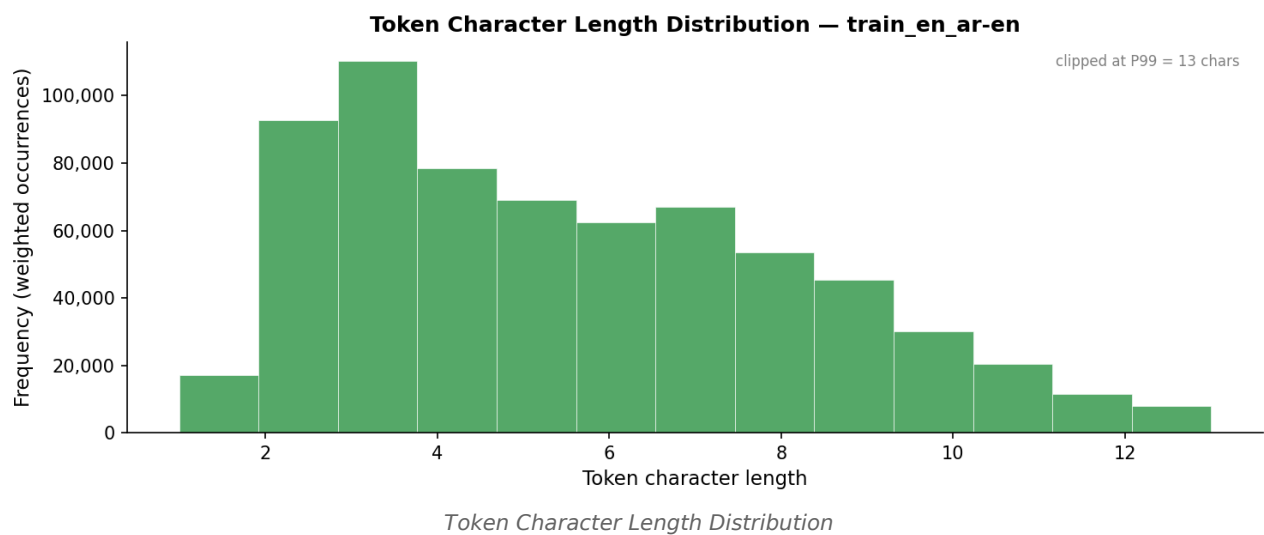




File: train_en_ar-en

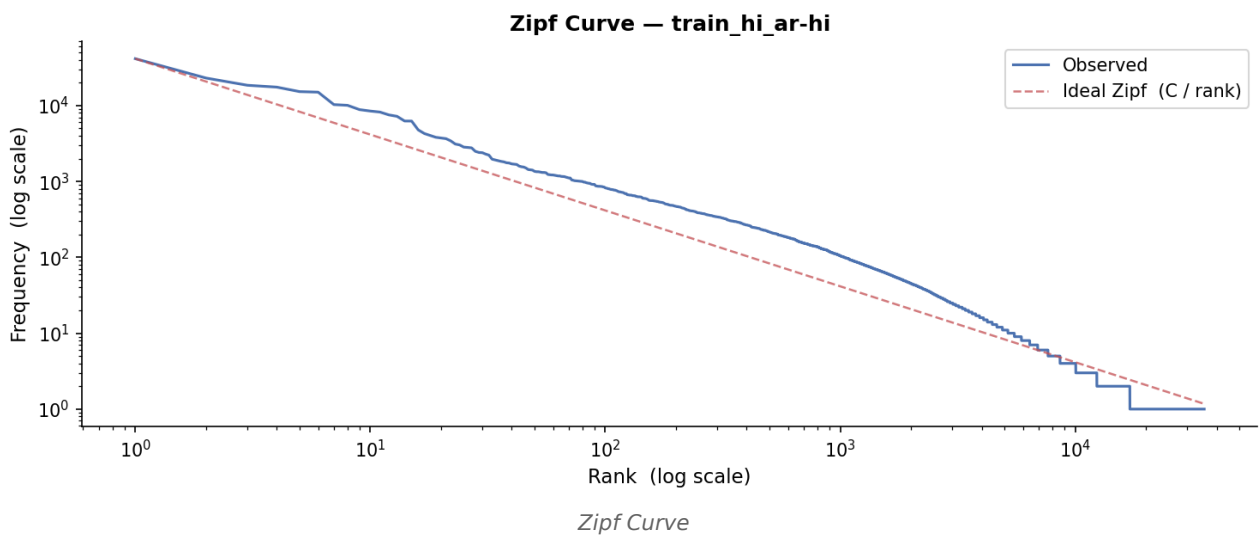
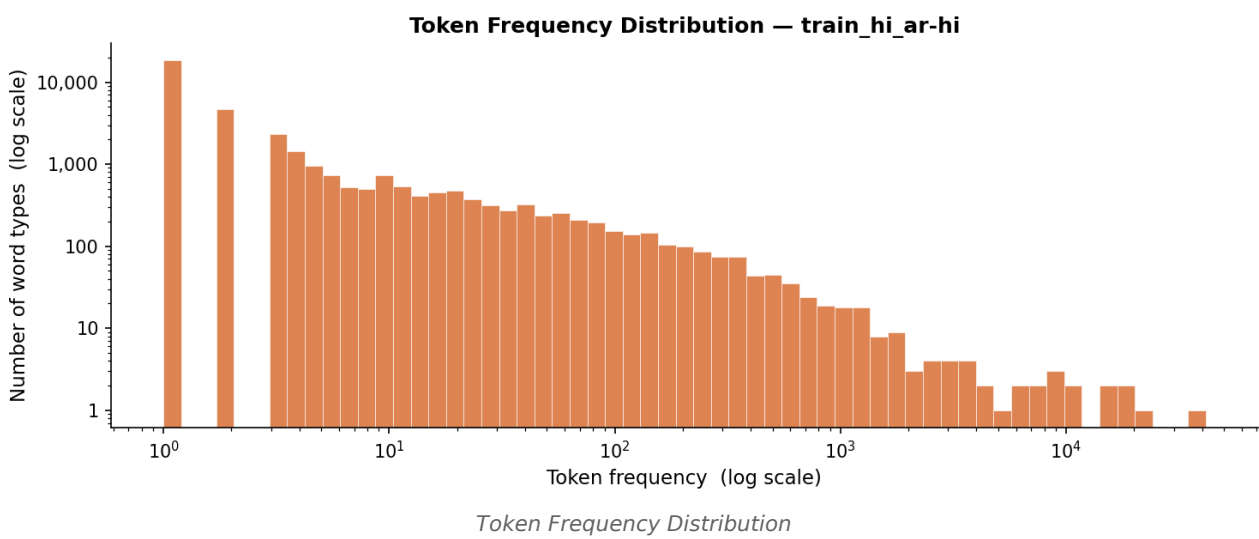
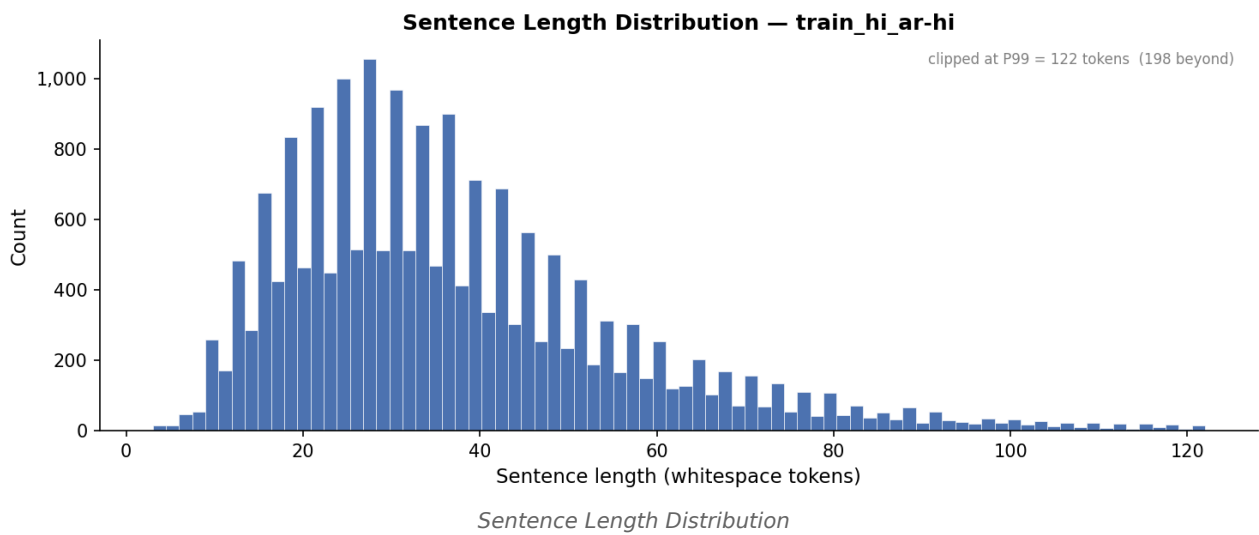
Total Sentences	21000
Unique Sentences	20994
Duplicate Sentences	6
Duplicate Pct	0.0286
Empty Sentences	0
Mean Len	31.9450
Median Len	28.0000
Min Len	3
Max Len	509
Std Len	19.0573
P95 Len	66.0000
P99 Len	101.0100
— Raw Vocab —	
Total Tokens	670845
Unique Tokens	58233
Ttr	0.0868
Freq Exactly 1	31205
Freq Exactly 2	8042
Freq 3 To 5	7961
Freq Over 100	749
Mean Token Char Len	5.5304
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125
— Norm Vocab —	
Total Tokens	670845
Unique Tokens	53325
Ttr	0.0795
Freq Exactly 1	28210
Freq Exactly 2	7304
Freq 3 To 5	7294
Freq Over 100	766
Mean Token Char Len	5.5304
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125

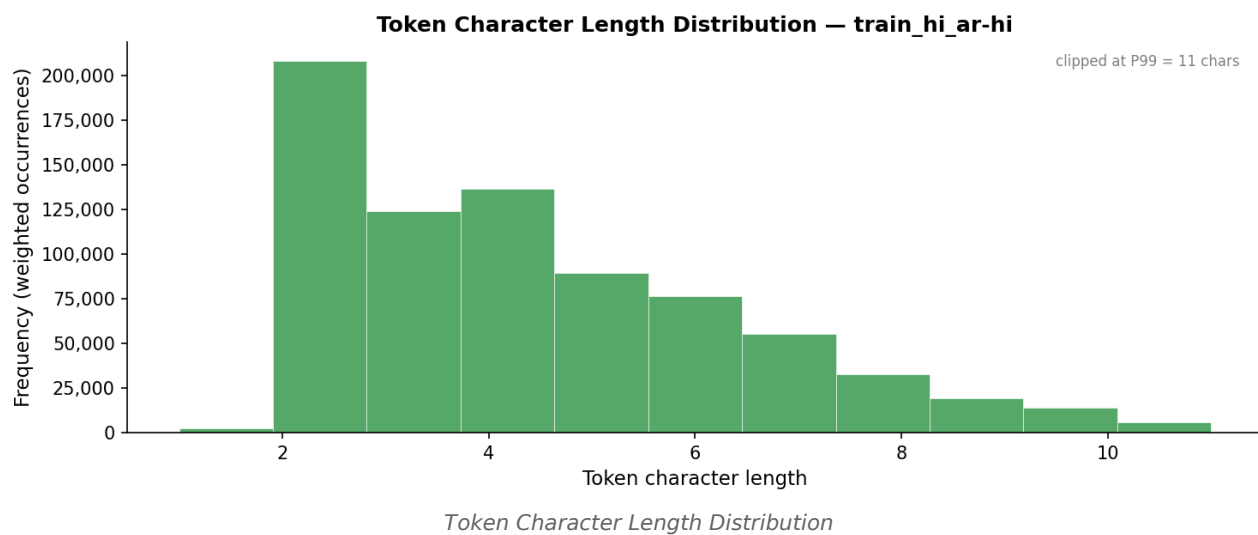




File: train_hi_ar-hi

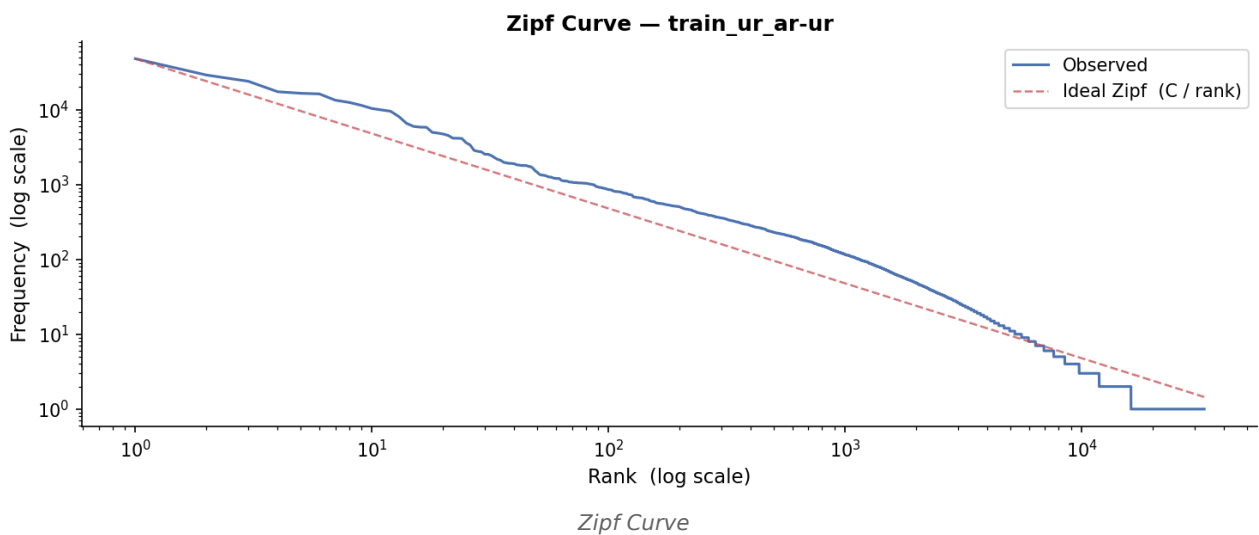
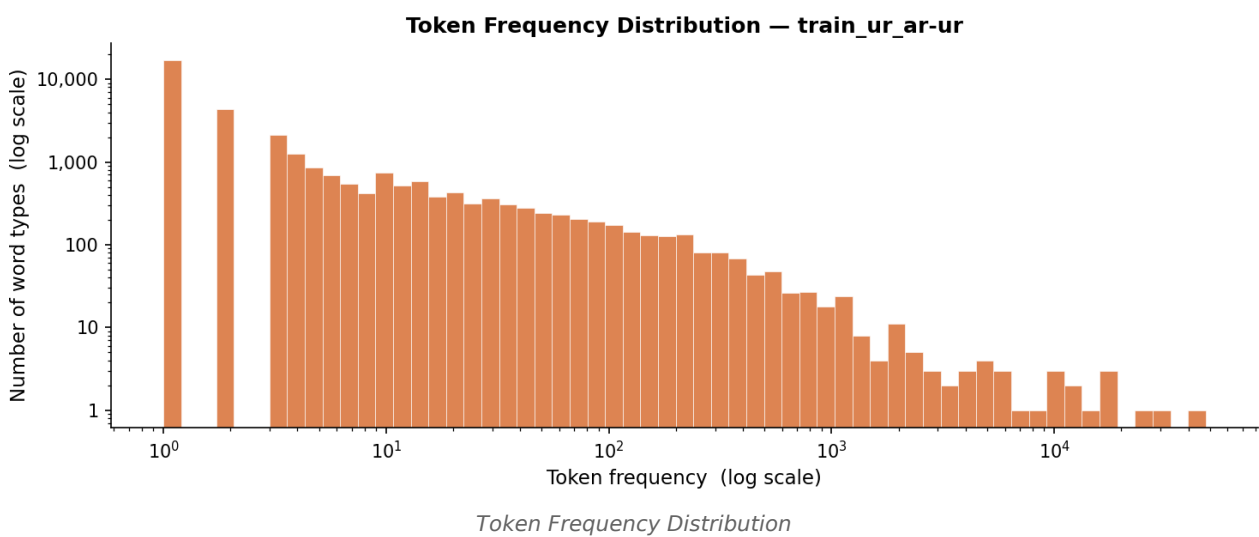
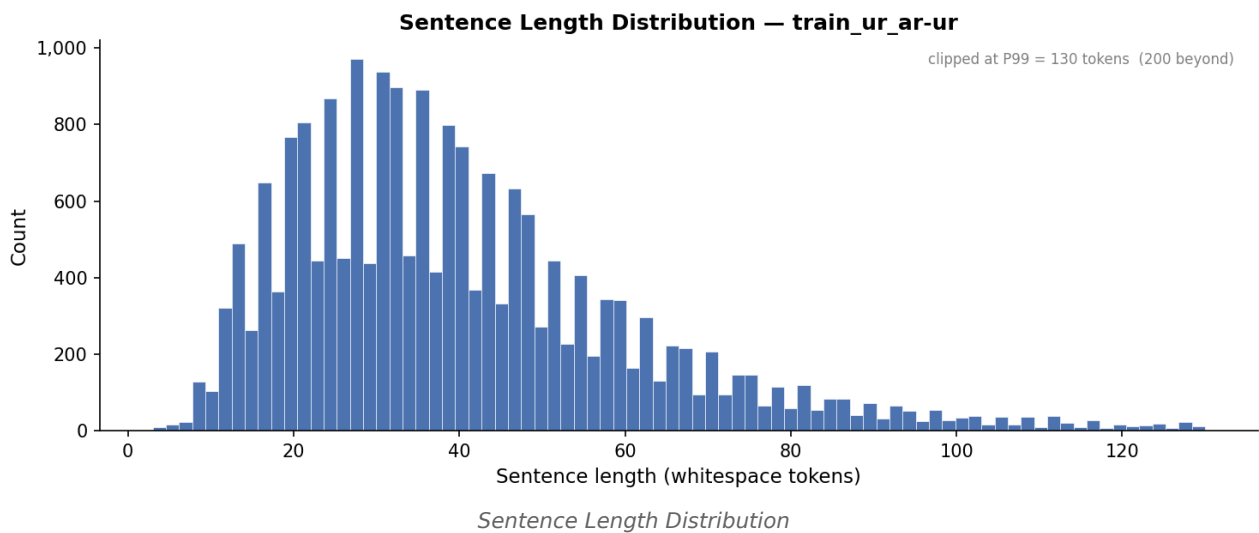
Total Sentences	20100
Unique Sentences	20091
Duplicate Sentences	9
Duplicate Pct	0.0448
Empty Sentences	0
Mean Len	38.3330
Median Len	33.0000
Min Len	3
Max Len	421
Std Len	22.6913
P95 Len	79.0500
P99 Len	122.0000
— Raw Vocab —	
Total Tokens	770493
Unique Tokens	35427
Ttr	0.0460
Freq Exactly 1	18345
Freq Exactly 2	4725
Freq 3 To 5	4700
Freq Over 100	1037
Mean Token Char Len	4.3637
Median Token Char Len	4.0000
Min Token Char Len	1
Max Token Char Len	49
— Norm Vocab —	
Total Tokens	770493
Unique Tokens	35426
Ttr	0.0460
Freq Exactly 1	18344
Freq Exactly 2	4725
Freq 3 To 5	4700
Freq Over 100	1037
Mean Token Char Len	4.3637
Median Token Char Len	4.0000
Min Token Char Len	1
Max Token Char Len	49

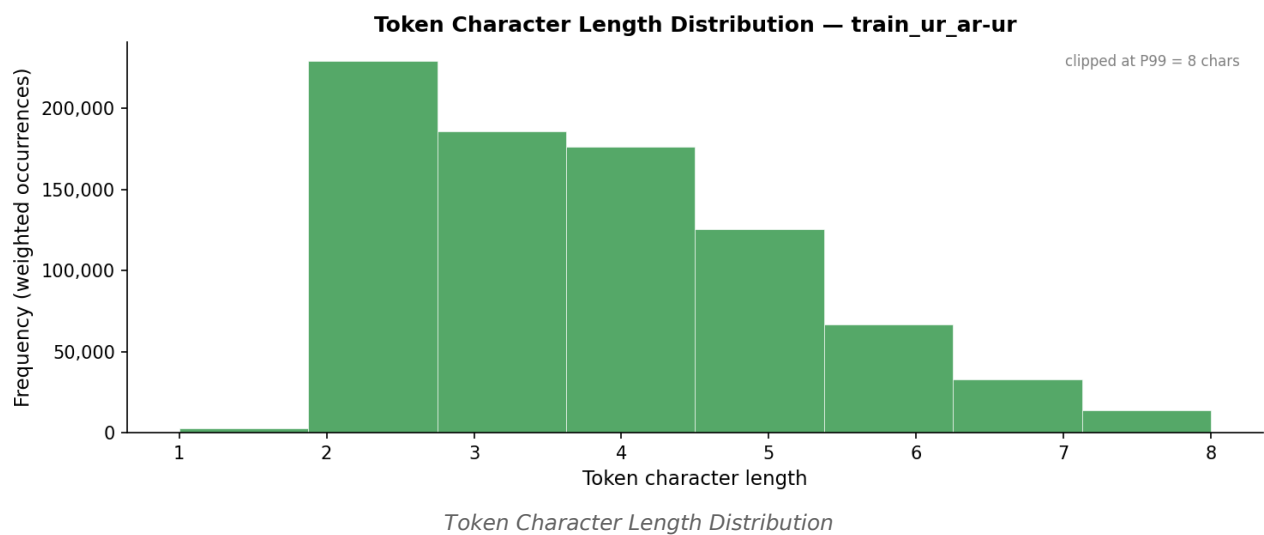




File: train_ur_ar-ur

Total Sentences	20300
Unique Sentences	20294
Duplicate Sentences	6
Duplicate Pct	0.0296
Empty Sentences	0
Mean Len	41.4184
Median Len	36.0000
Min Len	3
Max Len	547
Std Len	24.5862
P95 Len	86.0000
P99 Len	130.0000
— Raw Vocab —	
Total Tokens	840793
Unique Tokens	32960
Ttr	0.0392
Freq Exactly 1	16815
Freq Exactly 2	4298
Freq 3 To 5	4235
Freq Over 100	1143
Mean Token Char Len	3.7659
Median Token Char Len	4.0000
Min Token Char Len	1
Max Token Char Len	124
— Norm Vocab —	
Total Tokens	840793
Unique Tokens	32934
Ttr	0.0392
Freq Exactly 1	16796
Freq Exactly 2	4291
Freq 3 To 5	4233
Freq Over 100	1143
Mean Token Char Len	3.7659
Median Token Char Len	4.0000
Min Token Char Len	1
Max Token Char Len	124

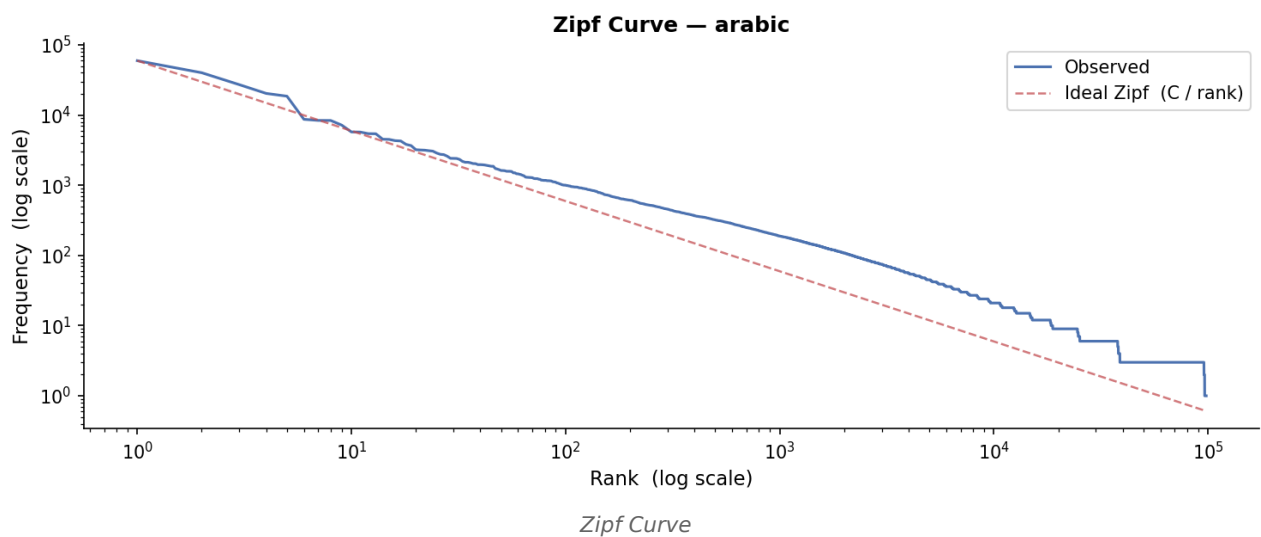
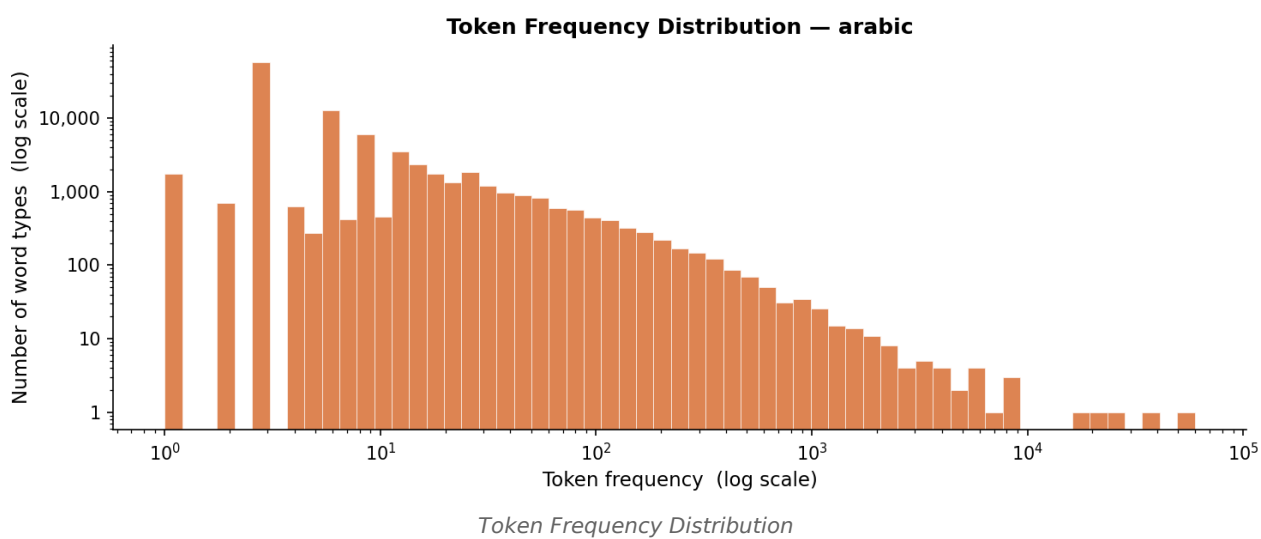
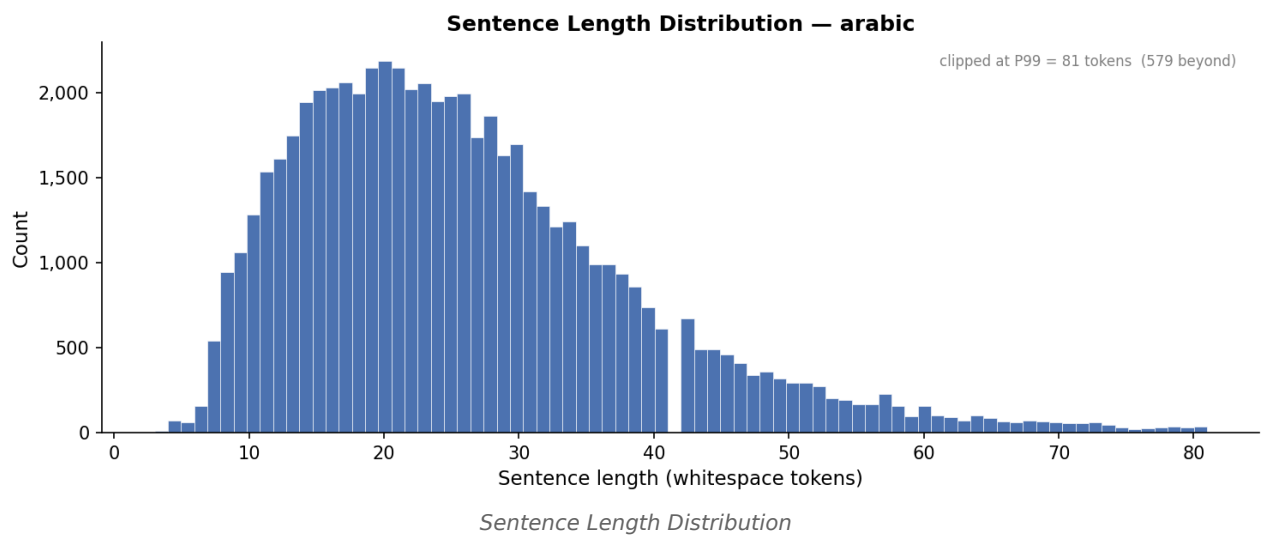


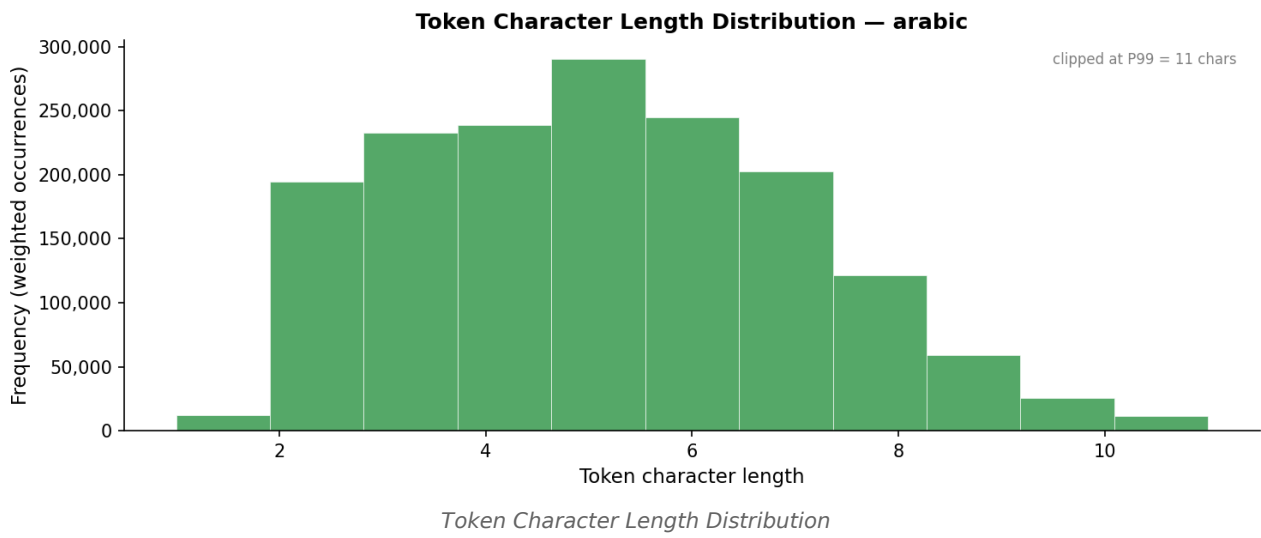


1.4 Language-Level Aggregation

Language: Arabic

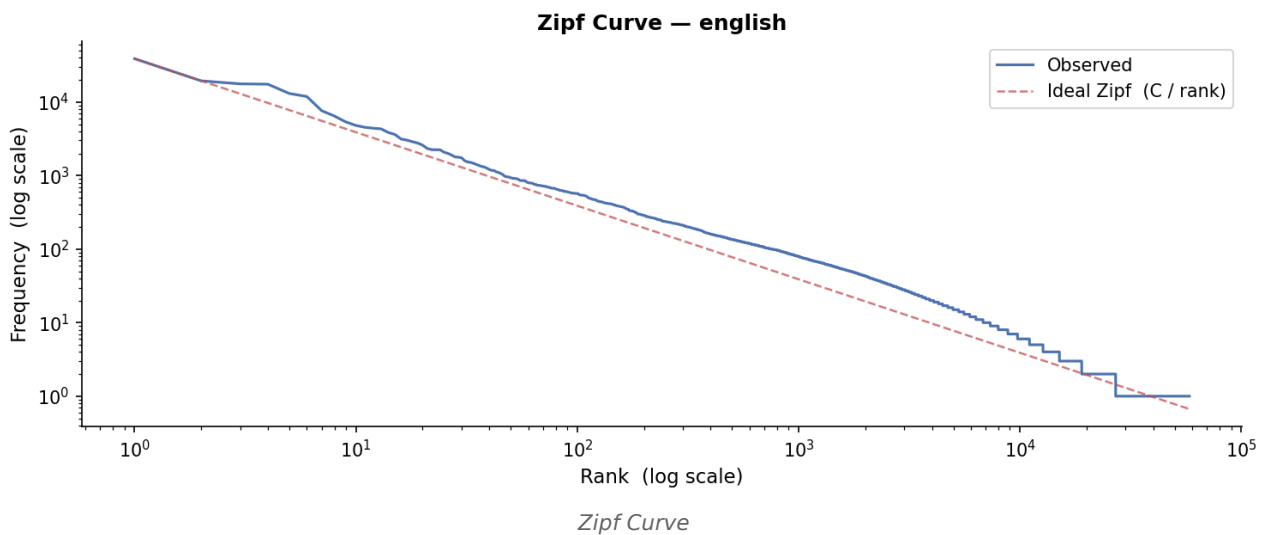
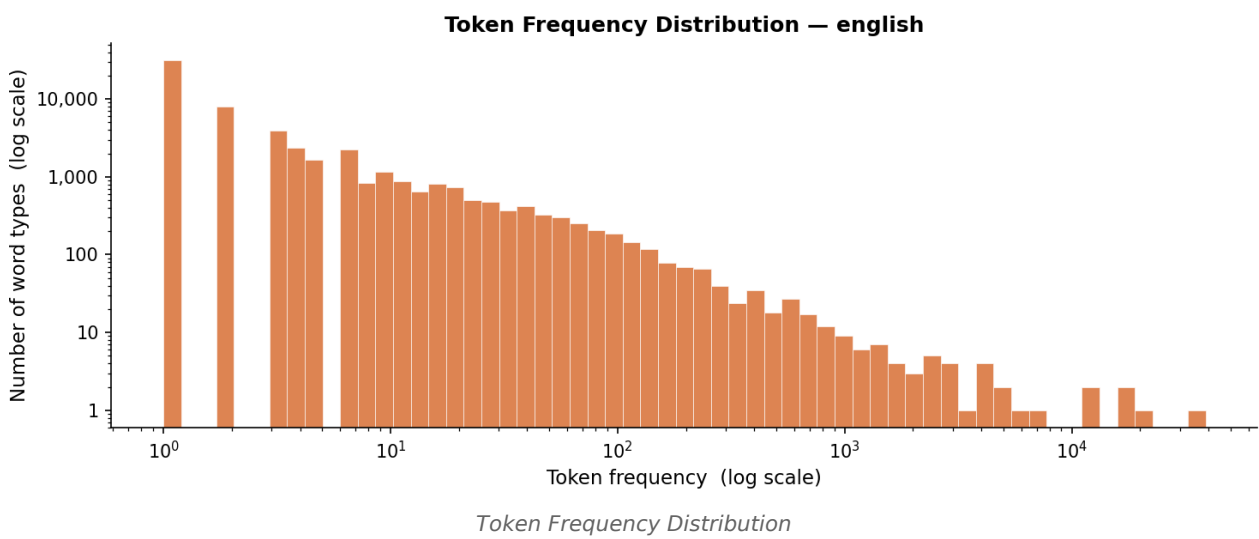
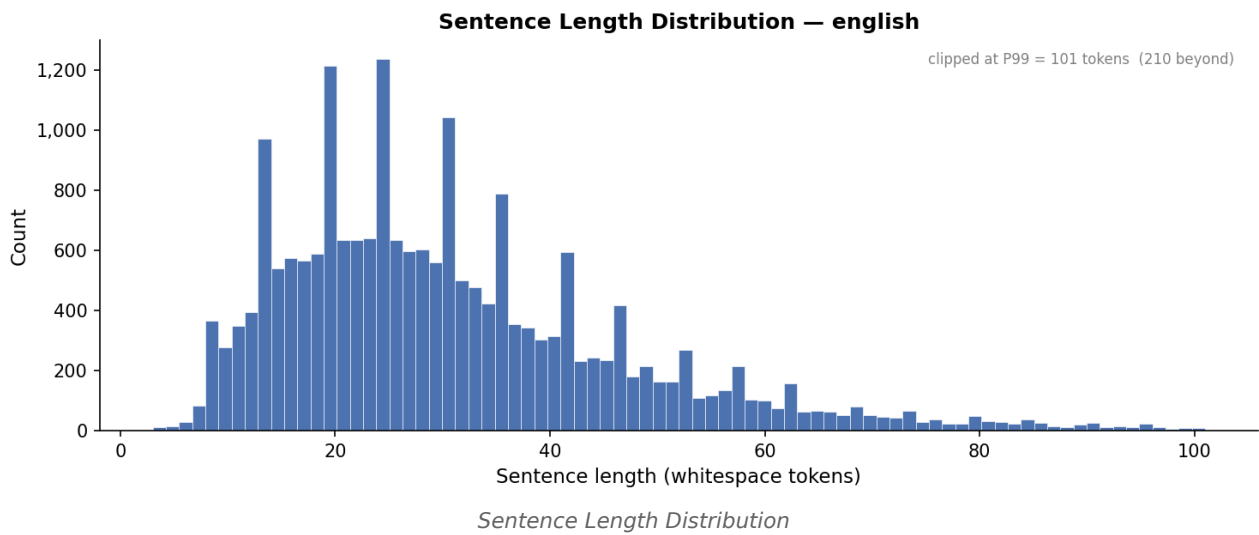
Total Sentences	61400
Unique Sentences	21086
Duplicate Sentences	40314
Duplicate Pct	65.6580
Empty Sentences	0
Mean Len	26.7350
Median Len	24.0000
Min Len	3
Max Len	120
Std Len	14.4783
P95 Len	53.0000
P99 Len	81.0000
— Raw Vocab —	
Total Tokens	1641528
Unique Tokens	98233
Ttr	0.0598
Freq Exactly 1	1735
Freq Exactly 2	710
Freq 3 To 5	57922
Freq Over 100	2166
Mean Token Char Len	5.1052
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125
— Norm Vocab —	
Total Tokens	1641528
Unique Tokens	98233
Ttr	0.0598
Freq Exactly 1	1735
Freq Exactly 2	710
Freq 3 To 5	57922
Freq Over 100	2166
Mean Token Char Len	5.1052
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125

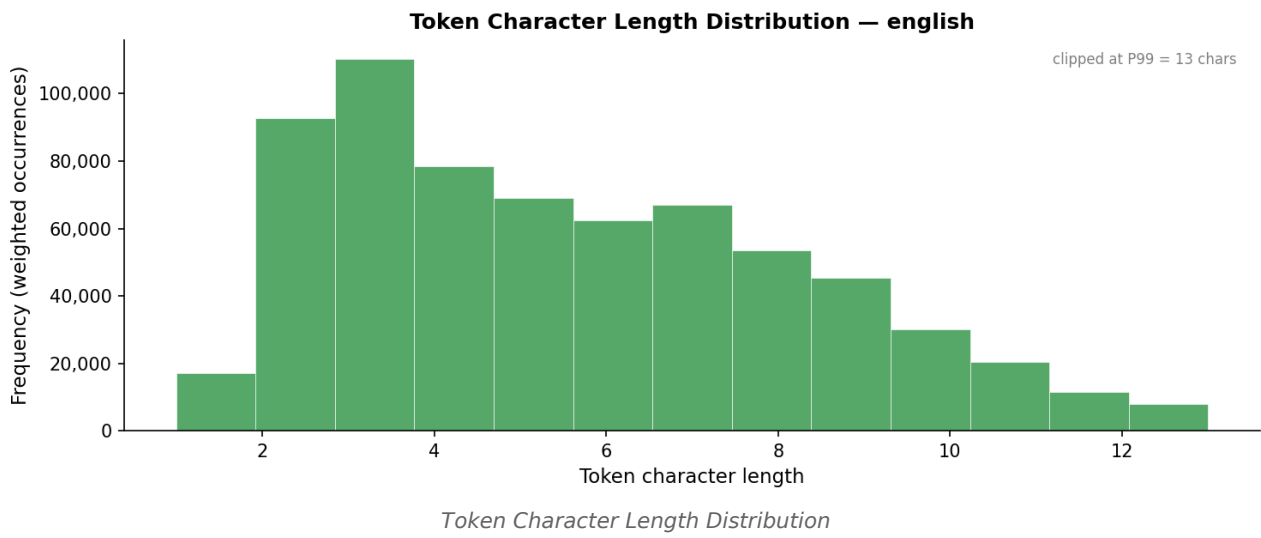




Language: English

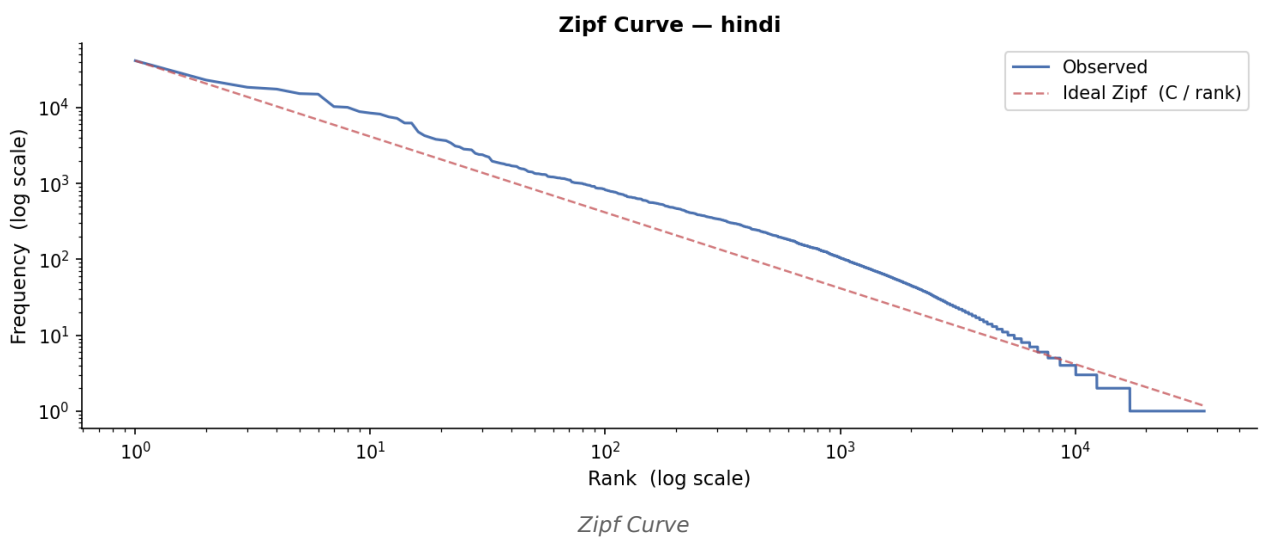
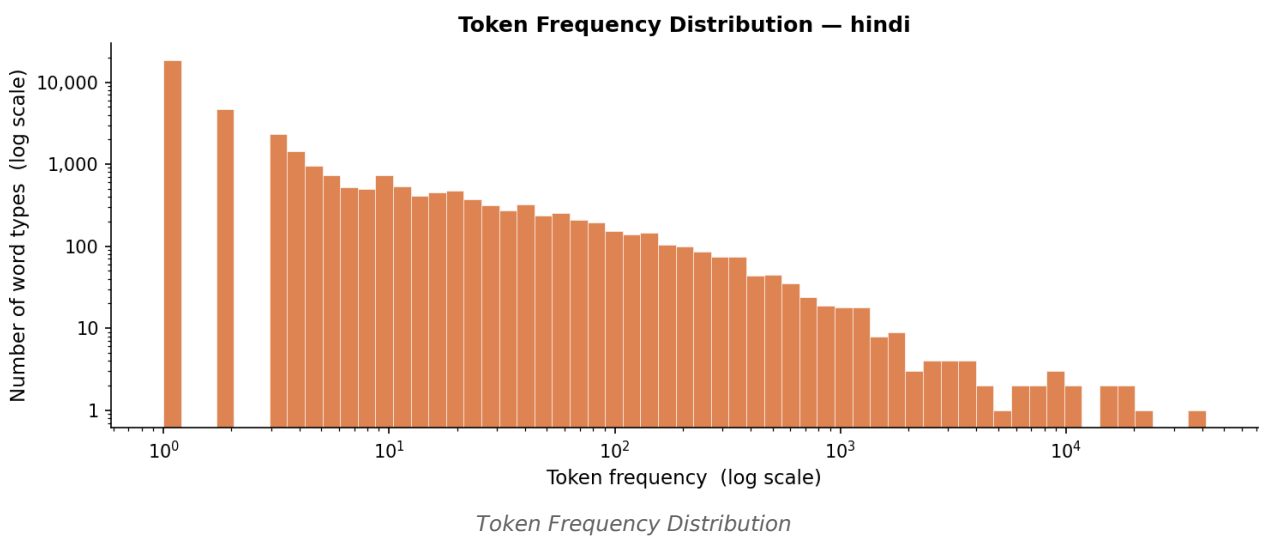
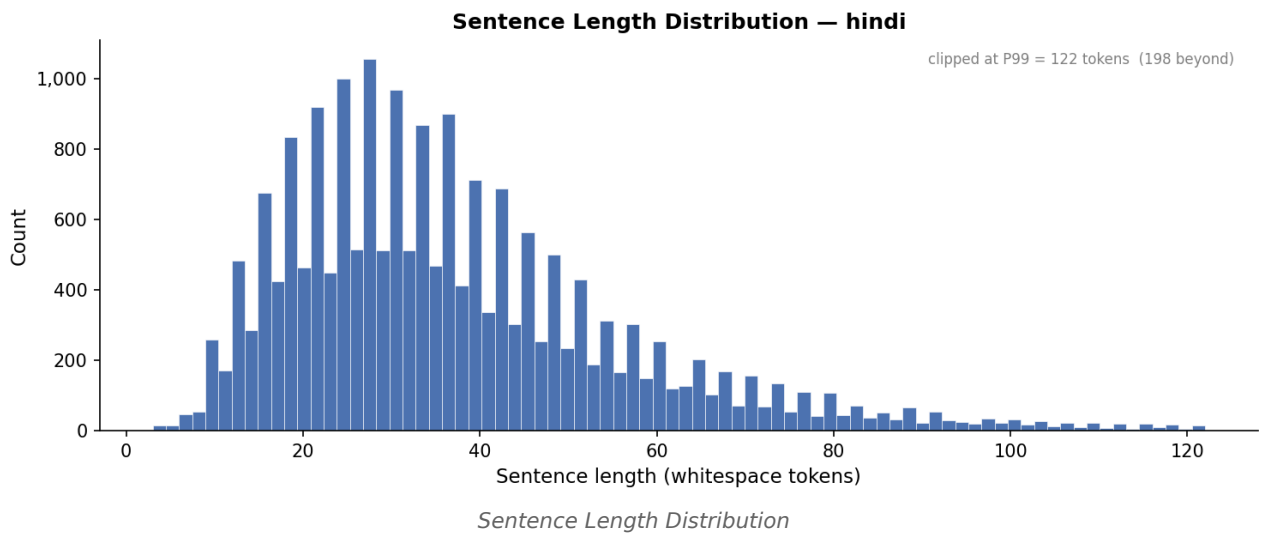
Total Sentences	21000
Unique Sentences	20994
Duplicate Sentences	6
Duplicate Pct	0.0286
Empty Sentences	0
Mean Len	31.9450
Median Len	28.0000
Min Len	3
Max Len	509
Std Len	19.0573
P95 Len	66.0000
P99 Len	101.0100
— Raw Vocab —	
Total Tokens	670845
Unique Tokens	58233
Ttr	0.0868
Freq Exactly 1	31205
Freq Exactly 2	8042
Freq 3 To 5	7961
Freq Over 100	749
Mean Token Char Len	5.5304
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125
— Norm Vocab —	
Total Tokens	670845
Unique Tokens	53325
Ttr	0.0795
Freq Exactly 1	28210
Freq Exactly 2	7304
Freq 3 To 5	7294
Freq Over 100	766
Mean Token Char Len	5.5304
Median Token Char Len	5.0000
Min Token Char Len	1
Max Token Char Len	125

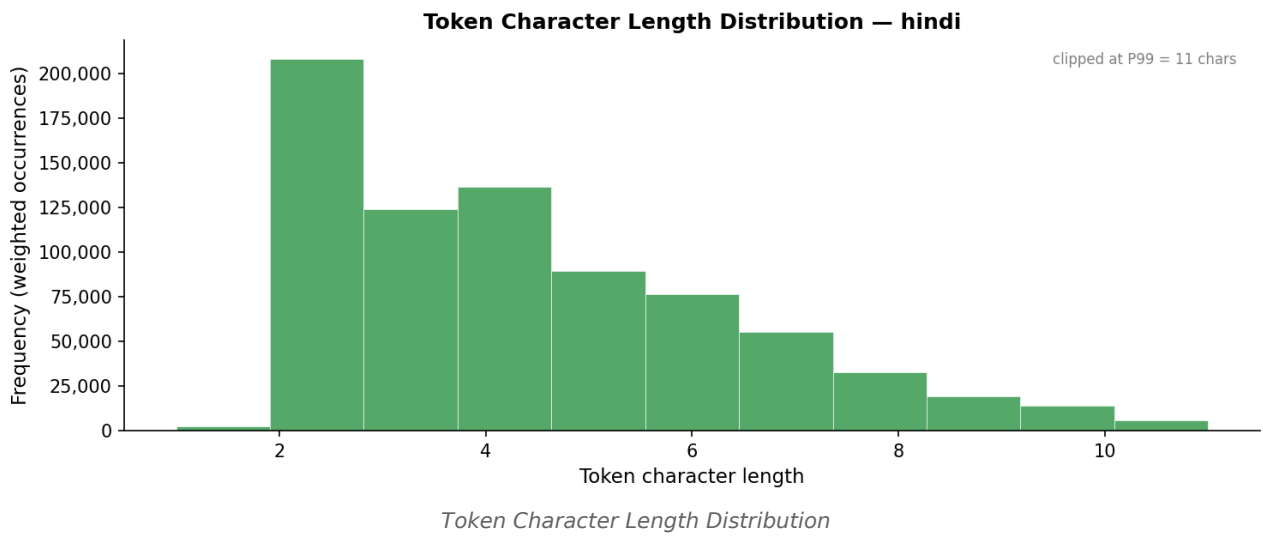




Language: Hindi

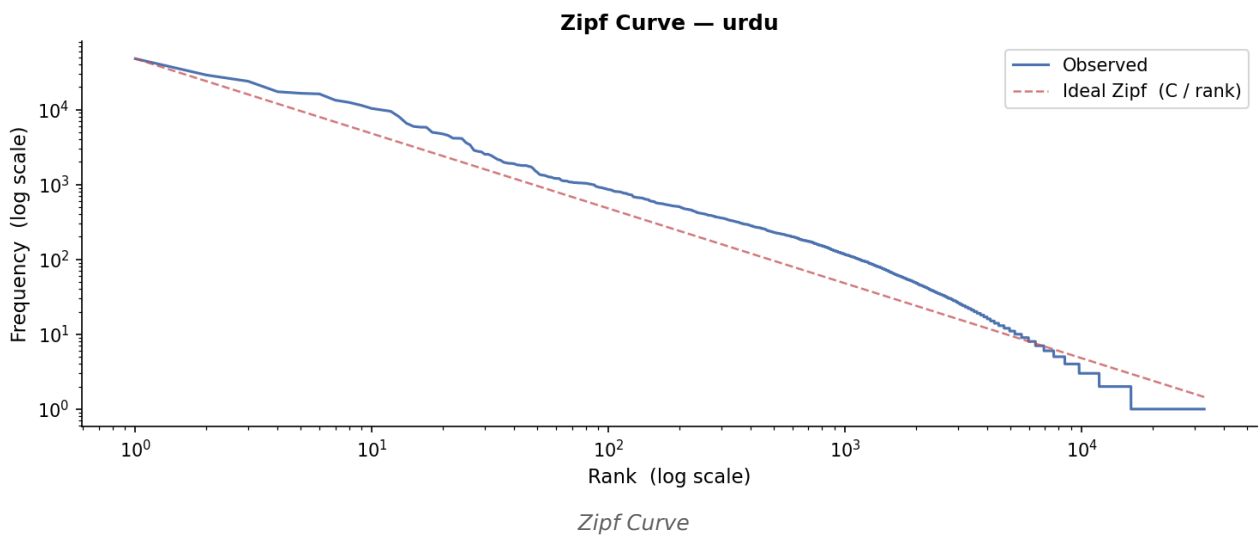
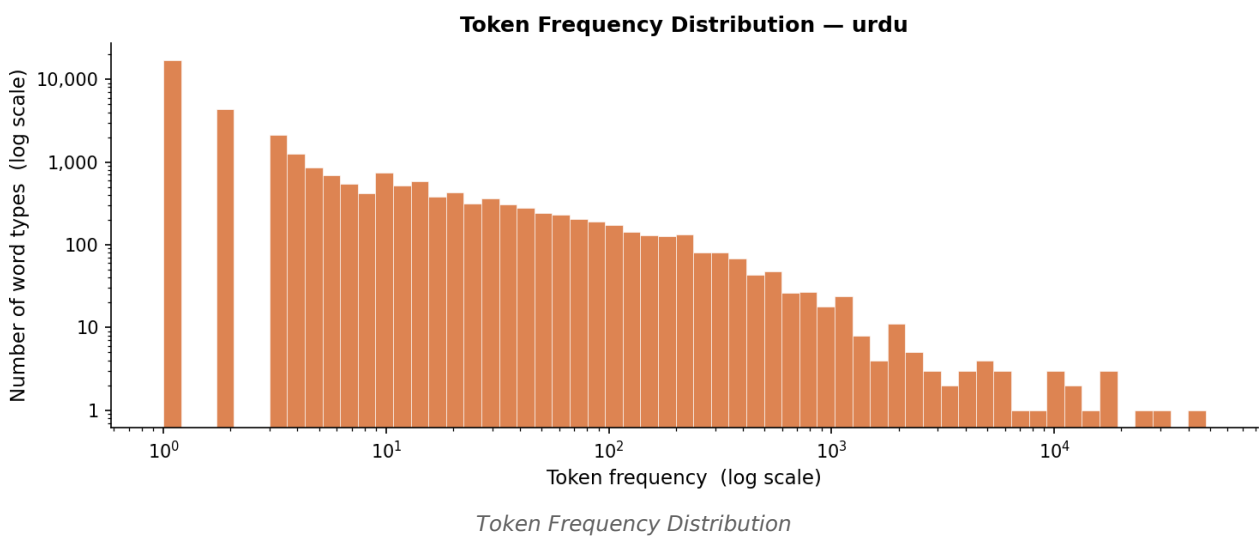
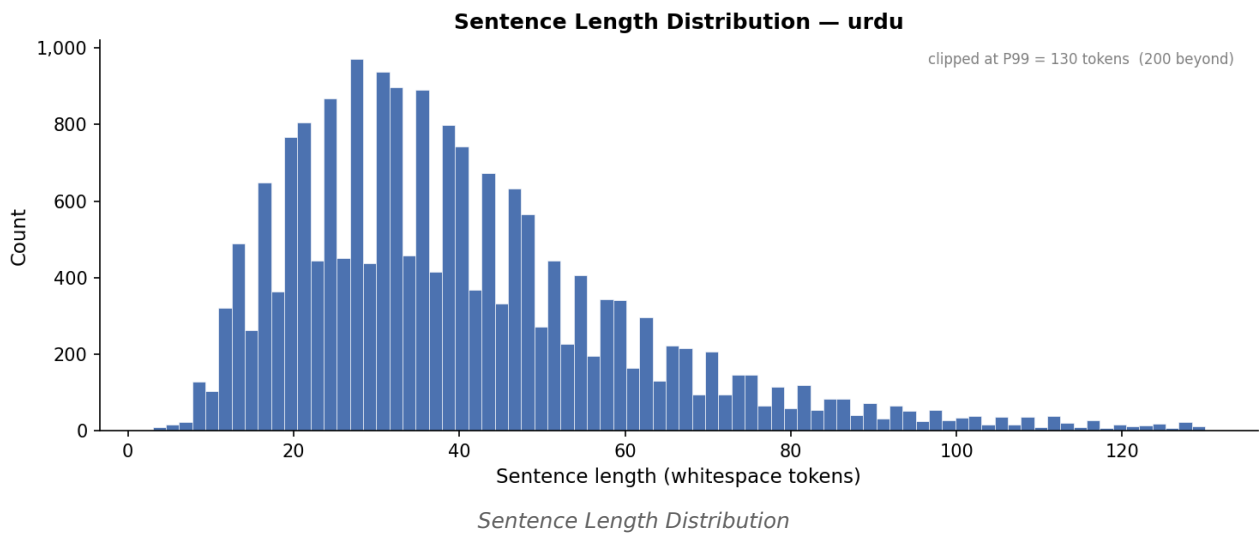
Total Sentences	20100
Unique Sentences	20091
Duplicate Sentences	9
Duplicate Pct	0.0448
Empty Sentences	0
Mean Len	38.3330
Median Len	33.0000
Min Len	3
Max Len	421
Std Len	22.6913
P95 Len	79.0500
P99 Len	122.0000
— Raw Vocab —	
Total Tokens	770493
Unique Tokens	35427
Ttr	0.0460
Freq Exactly 1	18345
Freq Exactly 2	4725
Freq 3 To 5	4700
Freq Over 100	1037
Mean Token Char Len	4.3637
Median Token Char Len	4.0000
Min Token Char Len	1
Max Token Char Len	49
— Norm Vocab —	
Total Tokens	770493
Unique Tokens	35426
Ttr	0.0460
Freq Exactly 1	18344
Freq Exactly 2	4725
Freq 3 To 5	4700
Freq Over 100	1037
Mean Token Char Len	4.3637
Median Token Char Len	4.0000
Min Token Char Len	1
Max Token Char Len	49

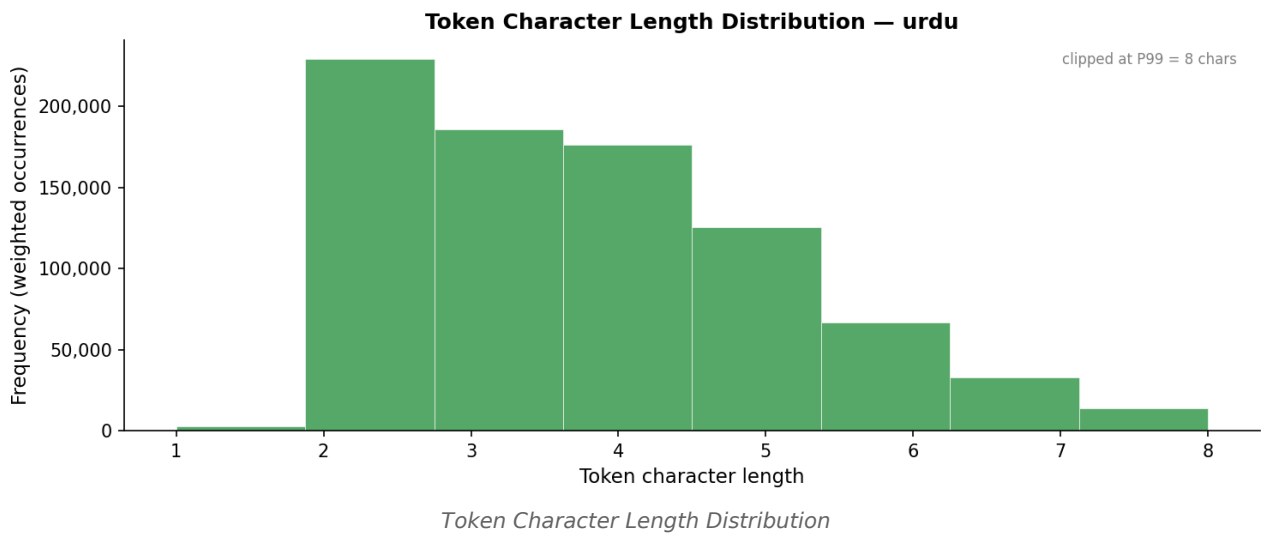




Language: Urdu

Total Sentences	20300
Unique Sentences	20294
Duplicate Sentences	6
Duplicate Pct	0.0296
Empty Sentences	0
Mean Len	41.4184
Median Len	36.0000
Min Len	3
Max Len	547
Std Len	24.5862
P95 Len	86.0000
P99 Len	130.0000
— Raw Vocab —	
Total Tokens	840793
Unique Tokens	32960
Ttr	0.0392
Freq Exactly 1	16815
Freq Exactly 2	4298
Freq 3 To 5	4235
Freq Over 100	1143
Mean Token Char Len	3.7659
Median Token Char Len	4.0000
Min Token Char Len	1
Max Token Char Len	124
— Norm Vocab —	
Total Tokens	840793
Unique Tokens	32934
Ttr	0.0392
Freq Exactly 1	16796
Freq Exactly 2	4291
Freq 3 To 5	4233
Freq Over 100	1143
Mean Token Char Len	3.7659
Median Token Char Len	4.0000
Min Token Char Len	1
Max Token Char Len	124





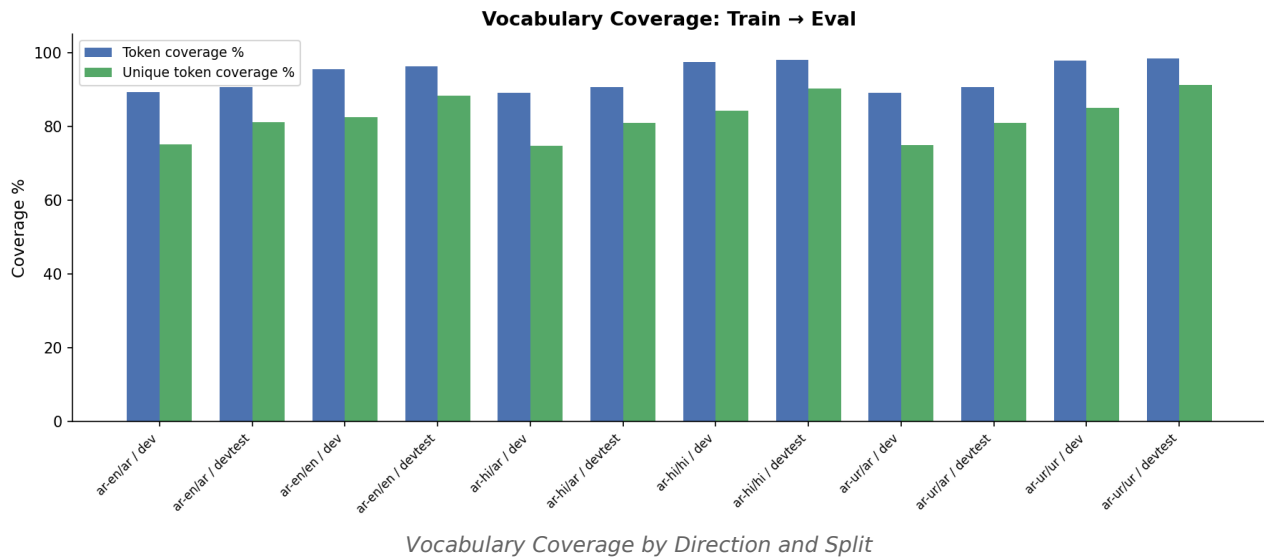
Part 2: Cross-Split Analysis

2.1 Dataset Statistics

direction	split	total_sente	unique_sen	duplicate_s	duplicate_p	total_token	unique_tok
ar-en/ar	train	21000	21000	0	0.0	558133	98062
ar-en/ar	dev	1000	1000	0	0.0	31817	12480
ar-en/ar	devtest	500	500	0	0.0	15885	6686
ar-en/en	train	21000	20994	6	0.0286	670845	58233
ar-en/en	dev	1000	999	1	0.1	39739	9203
ar-en/en	devtest	500	499	1	0.2	19608	5335
ar-hi/ar	train	20100	20100	0	0.0	539522	95932
ar-hi/ar	dev	1000	1000	0	0.0	31817	12480
ar-hi/ar	devtest	500	500	0	0.0	15885	6686
ar-hi/hi	train	20100	20091	9	0.0448	770493	35427
ar-hi/hi	dev	1000	999	1	0.1	46951	6367
ar-hi/hi	devtest	500	499	1	0.2	23351	3817
ar-ur/ar	train	20300	20300	0	0.0	543873	96320
ar-ur/ar	dev	1000	1000	0	0.0	31817	12480
ar-ur/ar	devtest	500	500	0	0.0	15868	6678
ar-ur/ur	train	20300	20294	6	0.0296	840793	32960
ar-ur/ur	dev	1000	999	1	0.1	51192	6293
ar-ur/ur	devtest	500	499	1	0.2	25268	3788

2.2 Vocabulary Coverage

direction	eval_split	token_coverage_pct	unique_token_cov_pct
ar-en/ar	dev	89.2039	75.0641
ar-en/ar	devtest	90.6956	81.1696
ar-en/en	dev	95.4302	82.484
ar-en/en	devtest	96.3586	88.2849
ar-hi/ar	dev	89.075	74.7917
ar-hi/ar	devtest	90.6012	80.9602
ar-hi/hi	dev	97.5208	84.2469
ar-hi/hi	devtest	98.0729	90.2017
ar-ur/ar	dev	89.1033	74.8638
ar-ur/ar	devtest	90.6352	81.0273
ar-ur/ur	dev	97.859	84.9198
ar-ur/ur	devtest	98.4407	91.2355



2.3 Out-of-Vocabulary (OOV) Analysis

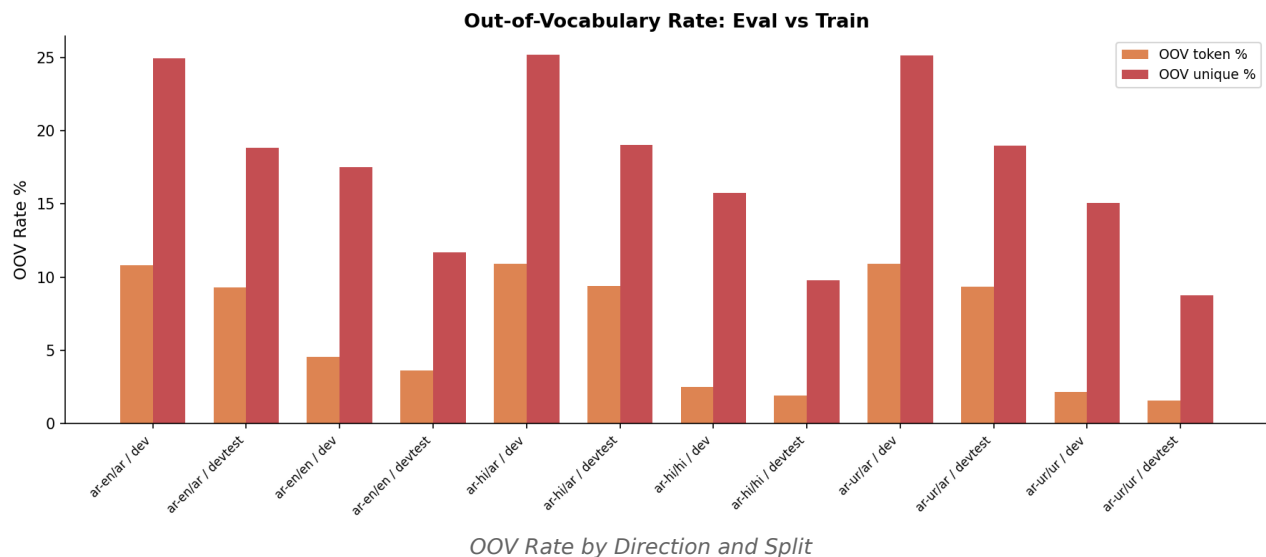
direction	eval_split	oov_token_cour	oov_token_pct	oov_unique_coi	oov_unique_pct
ar-en/ar	dev	3435	10.7961	3112	24.9359
ar-en/ar	devtest	1478	9.3044	1259	18.8304
ar-en/en	dev	1816	4.5698	1612	17.516
ar-en/en	devtest	714	3.6414	625	11.7151
ar-hi/ar	dev	3476	10.925	3146	25.2083
ar-hi/ar	devtest	1493	9.3988	1273	19.0398
ar-hi/hi	dev	1164	2.4792	1003	15.7531
ar-hi/hi	devtest	450	1.9271	374	9.7983
ar-ur/ar	dev	3467	10.8967	3137	25.1362
ar-ur/ar	devtest	1486	9.3648	1267	18.9727
ar-ur/ur	dev	1096	2.141	949	15.0802
ar-ur/ur	devtest	394	1.5593	332	8.7645

Top 50 OOV Tokens

direction	split	language	token	frequency
ar-en/ar	dev	ar	جنينة	16
ar-en/ar	dev	ar	سايكس	8
ar-en/ar	dev	ar	بيكو	7
ar-en/ar	dev	ar	القوة	6
ar-en/ar	dev	ar	أوكسيدنتال	5
ar-en/ar	dev	ar	والصغيرة	4
ar-en/ar	dev	ar	القبرصي	4
ar-en/ar	dev	ar	"إل.إن.جي	4
ar-en/ar	dev	ar	يؤخر	4
ar-en/ar	dev	ar	مصاها	4
ar-en/ar	dev	ar	"ريما	4
ar-en/ar	dev	ar	للتنمية	4
ar-en/ar	dev	ar	أوشن	4
ar-en/ar	dev	ar	"ماينور	4
ar-en/ar	dev	ar	"صن	4

ar-en/ar	dev	ar	رايز	Multilingual MT Dataset — EDA Report
ar-en/ar	dev	ar	عائمة	4
ar-en/ar	dev	ar	ومتناهية	3
ar-en/ar	dev	ar	الصغر	3
ar-en/ar	dev	ar	الموسوعة	3
ar-en/ar	dev	ar	وأبوظبي	3
ar-en/ar	dev	ar	للاستحباب	3
ar-en/ar	dev	ar	أمهل	3
ar-en/ar	dev	ar	درهم.	3
ar-en/ar	dev	ar	الأمد"	3
ar-en/ar	dev	ar	كلنا	3
ar-en/ar	dev	ar	المعاودة	3
ar-en/ar	dev	ar	سيارة	3
ar-en/ar	dev	ar	المفاوضين	3
ar-en/ar	dev	ar	كندا"	3
ar-en/ar	dev	ar	العفالق،	3
ar-en/ar	dev	ar	"الريتينا"	3
ar-en/ar	dev	ar	بتروليوم	3
ar-en/ar	dev	ar	العماني	3
ar-en/ar	dev	ar	أوسيمين	3
ar-en/ar	dev	ar	شخصيا،	3
ar-en/ar	dev	ar	"ميد	3
ar-en/ar	dev	ar	فندقا	3
ar-en/ar	dev	ar	مطحنة	3
ar-en/ar	dev	ar	ملحوظا،	3
ar-en/ar	dev	ar	المشغلة	3
ar-en/ar	dev	ar	الآجال	3
ar-en/ar	dev	ar	وشريحة	3
ar-en/ar	dev	ar	"عطرسية	3
ar-en/ar	dev	ar	التركي"	3
ar-en/ar	dev	ar	المكتسبات	3
ar-en/ar	dev	ar	حائل	3
ar-en/ar	dev	ar	"ويكيبيديا"	3
ar-en/ar	dev	ar	للبنشرة	3
ar-en/ar	dev	ar	"نيوز	3

... 2350 more rows not shown



2.4 Exact Duplicate / Leakage Detection

direction	comparison	identical_sents	duplicate_pct
ar-en/ar	train↔dev	0	0.0
ar-en/ar	train↔devtest	0	0.0
ar-en/en	train↔dev	1	0.1001
ar-en/en	train↔devtest	0	0.0
ar-hi/ar	train↔dev	0	0.0
ar-hi/ar	train↔devtest	0	0.0
ar-hi/hi	train↔dev	1	0.1001
ar-hi/hi	train↔devtest	0	0.0
ar-ur/ar	train↔dev	0	0.0
ar-ur/ar	train↔devtest	0	0.0
ar-ur/ur	train↔dev	1	0.1001
ar-ur/ur	train↔devtest	0	0.0

Duplicate Sentences (first 50 rows)

direction	split	sentence	frequency
ar-en/en	dev	*Quoted from "Al-Masry Al	1
ar-hi/hi	dev	*"अल-मसरी अल-यूम" से उद्धृत।	1
ar-ur/ur	dev	المصري اليوم سے اقتباس۔	1

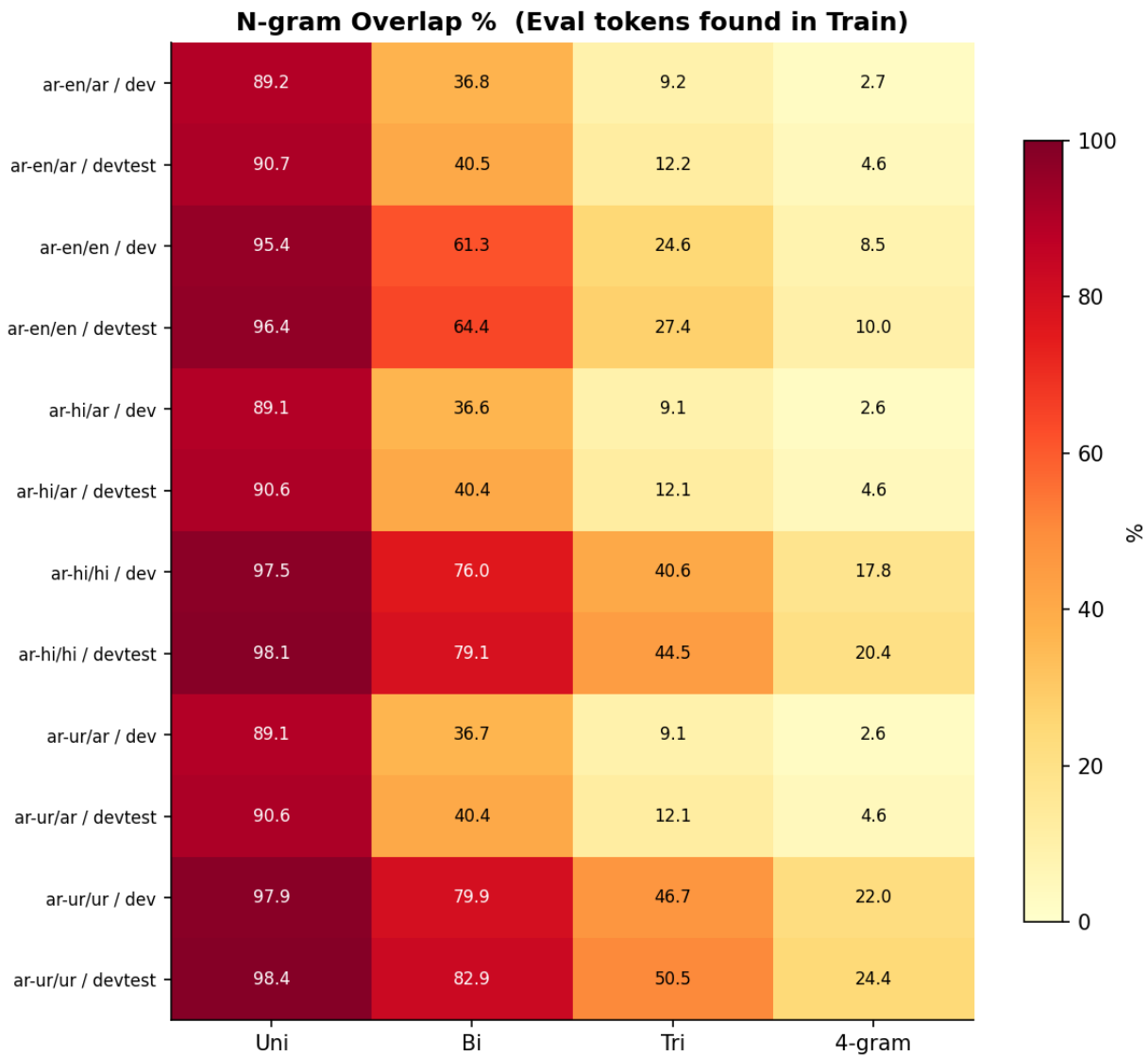
Leakage Risk Summary

Leakage Detection Report

*Auto-genera

2.5 N-Gram Overlap

direction	eval_split	unigram	bigram	trigram	fourgram
ar-en/ar	dev	89.2039	36.8238	9.1793	2.6651
ar-en/ar	devtest	90.6956	40.4615	12.1599	4.6159
ar-en/en	dev	95.4302	61.3387	24.6111	8.4542
ar-en/en	devtest	96.3586	64.4337	27.4344	10.0066
ar-hi/ar	dev	89.075	36.6162	9.0988	2.6408
ar-hi/ar	devtest	90.6012	40.377	12.1129	4.602
ar-hi/hi	dev	97.5208	76.0071	40.6042	17.7721
ar-hi/hi	devtest	98.0729	79.0687	44.4857	20.443
ar-ur/ar	dev	89.1033	36.6551	9.1223	2.6477
ar-ur/ar	devtest	90.6352	40.3956	12.1469	4.6075
ar-ur/ur	dev	97.859	79.9231	46.7393	22.0203
ar-ur/ur	devtest	98.4407	82.8811	50.5151	24.3941

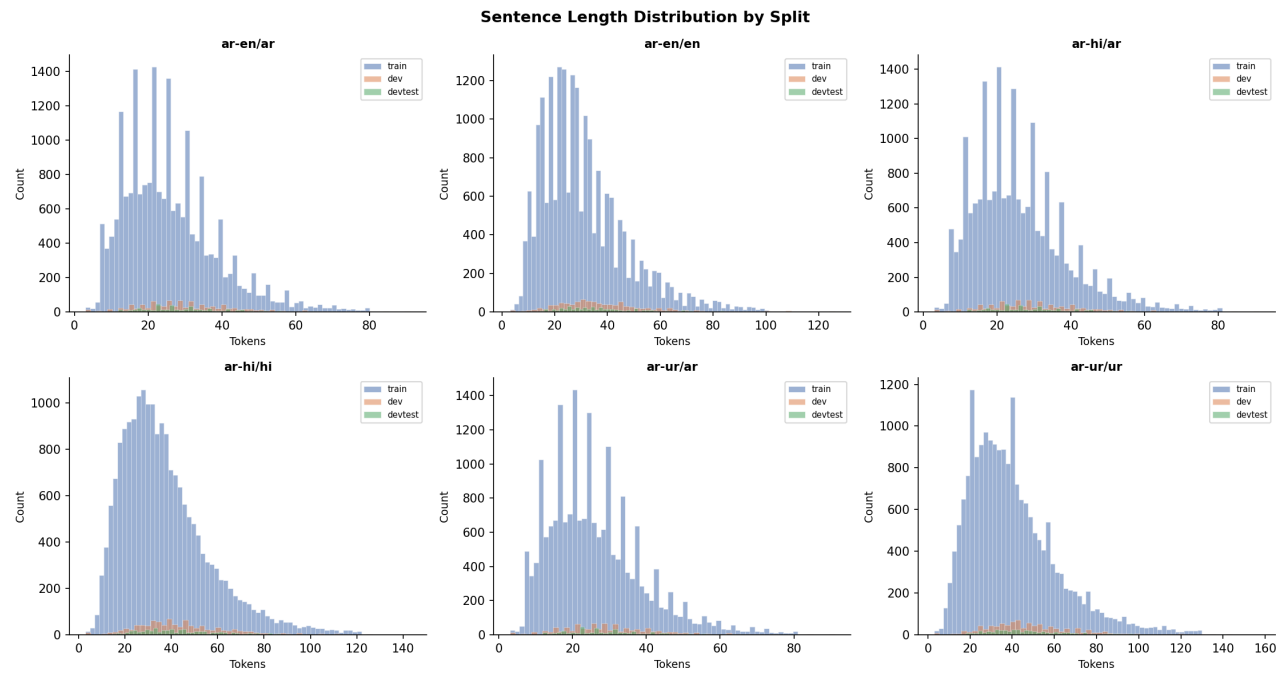


N-Gram Overlap Heatmap (Train vs Eval)

2.6 Sentence Length Distribution

direction	split	mean	median	std	p95	p99
ar-en/ar	train	26.5778	24.0	14.4019	52.0	80.0
ar-en/ar	dev	31.817	29.0	16.0298	62.0	91.03
ar-en/ar	devtest	31.77	29.0	15.2857	58.05	79.09
ar-en/en	train	31.945	28.0	19.0573	66.0	101.01
ar-en/en	dev	39.739	35.0	21.6469	76.0	126.0
ar-en/en	devtest	39.216	35.5	18.2284	72.05	93.0
ar-hi/ar	train	26.8419	24.0	14.531	53.0	81.0
ar-hi/ar	dev	31.817	29.0	16.0298	62.0	91.03
ar-hi/ar	devtest	31.77	29.0	15.2857	58.05	79.09
ar-hi/hi	train	38.333	33.0	22.6913	79.05	122.0
ar-hi/hi	dev	46.951	42.0	25.2052	90.0	143.04
ar-hi/hi	devtest	46.702	42.0	21.8504	88.05	110.02
ar-ur/ar	train	26.7918	24.0	14.5036	53.0	81.0
ar-ur/ar	dev	31.817	29.0	16.0298	62.0	91.03
ar-ur/ar	devtest	31.736	29.0	15.3152	58.05	79.09

ar-ur/ur	train	41.4184	36.0	24.5862	86.0	130.0
ar-ur/ur	dev	51.192	46.0	27.2039	97.05	157.05
ar-ur/ur	devtest	50.536	46.0	23.6456	96.0	123.07



Sentence Length: Train vs Dev vs DevTest

2.7 Vocabulary Frequency Shift

Direction: ar-en/ar

direction	token	train_freq	train_pct	dev_freq	dev_pct	devtest_freq	devtest_pct
ar-en/ar	في	20342	3.6447	1086	3.4133	542	3.412
ar-en/ar	من	13697	2.4541	661	2.0775	328	2.0648
ar-en/ar	على	9197	1.6478	545	1.7129	272	1.7123
ar-en/ar	إلى	6896	1.2355	469	1.4741	230	1.4479
ar-en/ar	أن	6255	1.1207	468	1.4709	229	1.4416
ar-en/ar	مع	2970	0.5321	152	0.4777	78	0.491
ar-en/ar	التي	2860	0.5124	143	0.4494	74	0.4658
ar-en/ar	عن	2849	0.5105	152	0.4777	76	0.4784
ar-en/ar	ما	2435	0.4363	149	0.4683	63	0.3966
ar-en/ar	هذه	1957	0.3506	101	0.3174	47	0.2959
ar-en/ar	أو	1944	0.3483	92	0.2892	50	0.3148
ar-en/ar	هذا	1848	0.3311	88	0.2766	33	0.2077
ar-en/ar	لا	1833	0.3284	57	0.1791	28	0.1763
ar-en/ar	الذي	1538	0.2756	94	0.2954	34	0.214
ar-en/ar	بعد	1536	0.2752	92	0.2892	34	0.214
ar-en/ar	خلال	1466	0.2627	118	0.3709	54	0.3399
ar-en/ar	بين	1452	0.2602	72	0.2263	35	0.2203
ar-en/ar	عام	1309	0.2345	42	0.132	39	0.2455
ar-en/ar	كما	1254	0.2247	104	0.3269	60	0.3777
ar-en/ar	,	1123	0.2012	2	0.0063	1	0.0063

Direction: ar-en/en

Multilingual MT Dataset — EDA Report

direction	token	train_freq	train_pct	dev_freq	dev_pct	devtest_freq	devtest_pct
ar-en/en	the	38876	5.7951	2553	6.4244	1261	6.431
ar-en/en	of	19410	2.8934	1133	2.8511	545	2.7795
ar-en/en	to	17682	2.6358	1085	2.7303	577	2.9427
ar-en/en	and	17490	2.6072	833	2.0962	513	2.6163
ar-en/en	in	13065	1.9475	732	1.842	330	1.683
ar-en/en	a	11887	1.7719	652	1.6407	312	1.5912
ar-en/en	that	7630	1.1374	549	1.3815	227	1.1577
ar-en/en	for	6442	0.9603	384	0.9663	223	1.1373
ar-en/en	The	5364	0.7996	373	0.9386	200	1.02
ar-en/en	on	4806	0.7164	289	0.7272	142	0.7242
ar-en/en	is	4527	0.6748	243	0.6115	118	0.6018
ar-en/en	by	4402	0.6562	328	0.8254	168	0.8568
ar-en/en	with	4315	0.6432	216	0.5435	96	0.4896
ar-en/en	from	3854	0.5745	231	0.5813	130	0.663
ar-en/en	as	3618	0.5393	213	0.536	83	0.4233
ar-en/en	their	3140	0.4681	189	0.4756	116	0.5916
ar-en/en	has	3034	0.4523	203	0.5108	102	0.5202
ar-en/en	its	2888	0.4305	218	0.5486	94	0.4794
ar-en/en	at	2771	0.4131	203	0.5108	83	0.4233
ar-en/en	was	2594	0.3867	154	0.3875	50	0.255

Direction: ar-hi/ar

direction	token	train_freq	train_pct	dev_freq	dev_pct	devtest_freq	devtest_pct
ar-hi/ar	في	19510	3.6162	1086	3.4133	542	3.412
ar-hi/ar	من	13130	2.4336	661	2.0775	328	2.0648
ar-hi/ar	على	8894	1.6485	545	1.7129	272	1.7123
ar-hi/ar	إلى	6693	1.2405	469	1.4741	230	1.4479
ar-hi/ar	أن	6134	1.1369	468	1.4709	229	1.4416
ar-hi/ar	مع	2845	0.5273	152	0.4777	78	0.491
ar-hi/ar	عن	2763	0.5121	152	0.4777	76	0.4784
ar-hi/ar	التي	2740	0.5079	143	0.4494	74	0.4658
ar-hi/ar	ما	2373	0.4398	149	0.4683	63	0.3966
ar-hi/ar	هذه	1886	0.3496	101	0.3174	47	0.2959
ar-hi/ar	أو	1882	0.3488	92	0.2892	50	0.3148
ar-hi/ar	هذا	1783	0.3305	88	0.2766	33	0.2077
ar-hi/ar	لا	1782	0.3303	57	0.1791	28	0.1763
ar-hi/ar	بعد	1495	0.2771	92	0.2892	34	0.214
ar-hi/ar	الذي	1471	0.2726	94	0.2954	34	0.214
ar-hi/ar	خلال	1420	0.2632	118	0.3709	54	0.3399
ar-hi/ar	بين	1406	0.2606	72	0.2263	35	0.2203
ar-hi/ar	عام	1252	0.2321	42	0.132	39	0.2455
ar-hi/ar	كما	1207	0.2237	104	0.3269	60	0.3777
ar-hi/ar	غير	1052	0.195	47	0.1477	39	0.2455

Direction: ar-hi/hi

direction	token	train_freq	train_pct	dev_freq	dev_pct	devtest_freq	devtest_pct
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ar-hi/hi	के	41501	5.3863	2650	5.6442	1332	5.7043
ar-hi/hi	मे	23066	2.9937	1425	3.0351	666	2.8521
ar-hi/hi	की	18524	2.4042	1245	2.6517	593	2.5395
ar-hi/hi	और	17507	2.2722	862	1.836	535	2.2911
ar-hi/hi	को	15280	1.9831	898	1.9126	465	1.9913
ar-hi/hi	से	15028	1.9504	891	1.8977	464	1.9871
ar-hi/hi	का	10303	1.3372	607	1.2928	315	1.349
ar-hi/hi	है।	10081	1.3084	644	1.3716	317	1.3575
ar-hi/hi	पर	8846	1.1481	620	1.3205	308	1.319
ar-hi/hi	ने	8472	1.0996	598	1.2737	322	1.379
ar-hi/hi	एक	8210	1.0656	411	0.8754	173	0.7409
ar-hi/hi	कि	7535	0.9779	581	1.2375	247	1.0578
ar-hi/hi	लिए	7217	0.9367	388	0.8264	248	1.0621
ar-hi/hi	है,	6259	0.8123	418	0.8903	213	0.9122
ar-hi/hi	करने	6248	0.8109	385	0.82	220	0.9421
ar-hi/hi	है	4781	0.6205	274	0.5836	126	0.5396
ar-hi/hi	जो	4277	0.5551	250	0.5325	113	0.4839
ar-hi/hi	यह	4036	0.5238	247	0.5261	85	0.364
ar-hi/hi	इस	3820	0.4958	319	0.6794	143	0.6124
ar-hi/hi	रूप	3737	0.485	212	0.4515	107	0.4582

Direction: ar-ur/ar

direction	token	train_freq	train_pct	dev_freq	dev_pct	devtest_freq	devtest_pct
ar-ur/ar	في	19692	3.6207	1086	3.4133	542	3.4157
ar-ur/ar	من	13256	2.4373	661	2.0775	328	2.0671
ar-ur/ar	على	8982	1.6515	545	1.7129	273	1.7204
ar-ur/ar	إلى	6750	1.2411	469	1.4741	230	1.4495
ar-ur/ar	أن	6186	1.1374	468	1.4709	229	1.4432
ar-ur/ar	مع	2884	0.5303	152	0.4777	78	0.4916
ar-ur/ar	عن	2779	0.511	152	0.4777	75	0.4726
ar-ur/ar	التي	2773	0.5099	143	0.4494	74	0.4663
ar-ur/ar	ما	2395	0.4404	149	0.4683	63	0.397
ar-ur/ar	هذه	1904	0.3501	101	0.3174	47	0.2962
ar-ur/ar	أو	1898	0.349	92	0.2892	50	0.3151
ar-ur/ar	هذا	1801	0.3311	88	0.2766	33	0.208
ar-ur/ar	لا	1798	0.3306	57	0.1791	28	0.1765
ar-ur/ar	بعد	1498	0.2754	92	0.2892	34	0.2143
ar-ur/ar	الذي	1490	0.274	94	0.2954	34	0.2143
ar-ur/ar	خلال	1434	0.2637	118	0.3709	54	0.3403
ar-ur/ar	بين	1413	0.2598	72	0.2263	35	0.2206
ar-ur/ar	عام	1259	0.2315	42	0.132	39	0.2458
ar-ur/ar	كما	1212	0.2228	104	0.3269	60	0.3781
ar-ur/ar	غير	1067	0.1962	47	0.1477	40	0.2521

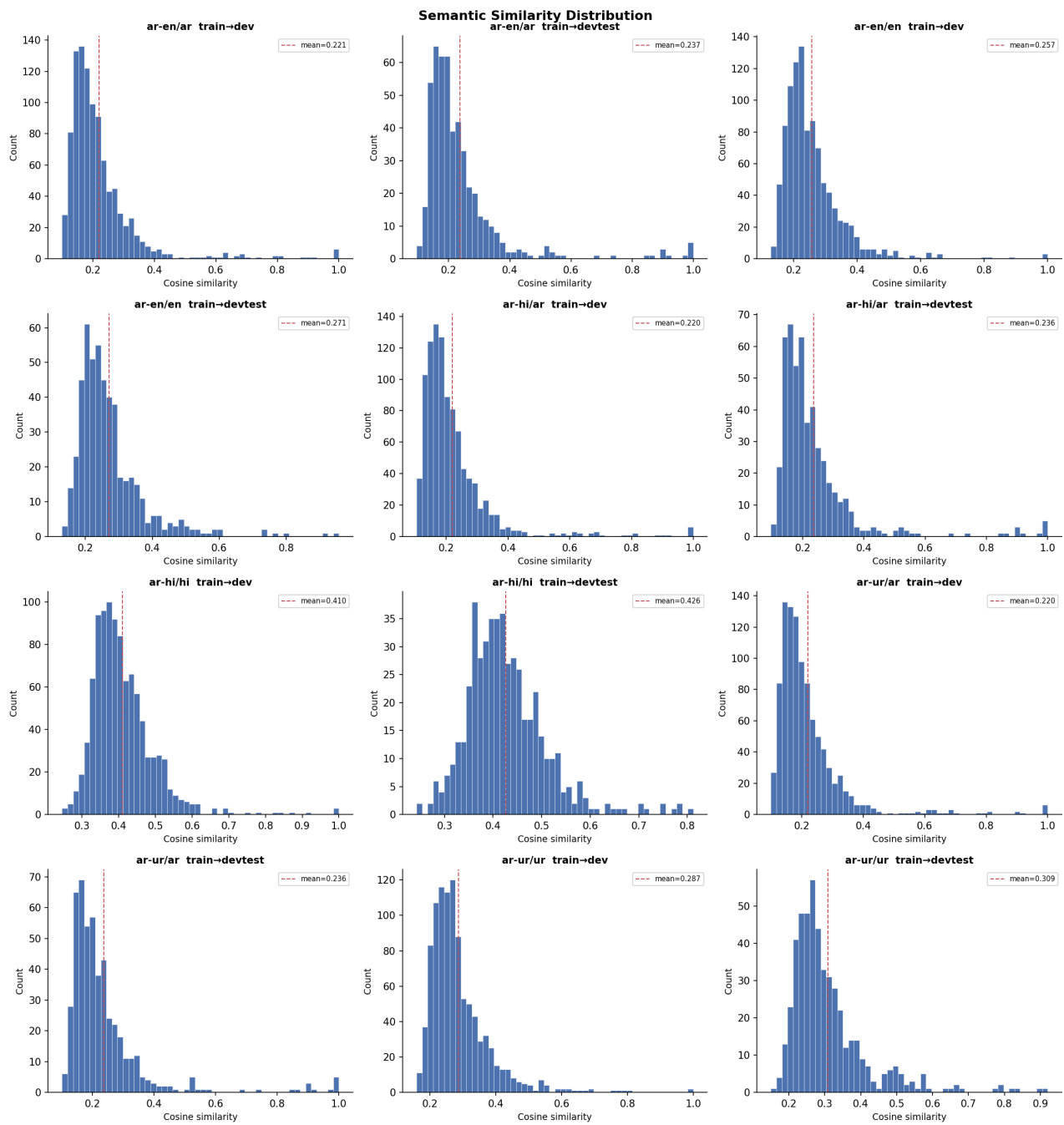
Direction: ar-ur/ur

direction	token	train_freq	train_pct	dev_freq	dev_pct	devtest_freq	devtest_pct
ar-ur/ur	ک	47768	5.6813	2991	5.8427	1546	6.1184
ar-ur/ur	کی	28877	3.4345	1929	3.7682	911	3.6054
ar-ur/ur	میں	23860	2.8378	1441	2.8149	682	2.6991

ar-ur/ur	اور	17284	2.0557	845	1.6506	517	2.0461
ar-ur/ur	س	16489	1.9611	1003	1.9593	471	1.864
ar-ur/ur	کو	16164	1.9225	926	1.8089	498	1.9709
ar-ur/ur	پر	13335	1.586	869	1.6975	442	1.7492
ar-ur/ur	ن	12458	1.4817	852	1.6643	448	1.773
ar-ur/ur	کا	11401	1.356	674	1.3166	341	1.3495
ar-ur/ur	اس	10325	1.228	669	1.3068	307	1.215
ar-ur/ur	ہے	9877	1.1747	643	1.2561	309	1.2229
ar-ur/ur	ک	9483	1.1279	708	1.383	308	1.2189
ar-ur/ur	ایک	8053	0.9578	389	0.7599	157	0.6213
ar-ur/ur	لی	6550	0.779	367	0.7169	253	1.0013
ar-ur/ur	ہے،	5953	0.708	399	0.7794	194	0.7678
ar-ur/ur		5820	0.6922	344	0.672	152	0.6016
ar-ur/ur	کر،	5793	0.689	326	0.6368	190	0.7519
ar-ur/ur	ی	4957	0.5896	306	0.5977	107	0.4235
ar-ur/ur	کیا	4848	0.5766	321	0.6271	130	0.5145
ar-ur/ur	جو	4712	0.5604	285	0.5567	127	0.5026

2.8 Semantic Similarity (TF-IDF)

direction	eval_split	method	mean_similarity	median_similarity	p95_similarity
ar-en/ar	dev	tfidf	0.2208	0.1901	0.3834
ar-en/ar	devtest	tfidf	0.237	0.1995	0.4429
ar-en/en	dev	tfidf	0.257	0.2318	0.4088
ar-en/en	devtest	tfidf	0.271	0.2439	0.4721
ar-hi/ar	dev	tfidf	0.2202	0.1892	0.3822
ar-hi/ar	devtest	tfidf	0.2359	0.1989	0.4442
ar-hi/hi	dev	tfidf	0.4096	0.3941	0.5455
ar-hi/hi	devtest	tfidf	0.4264	0.4145	0.5726
ar-ur/ar	dev	tfidf	0.2202	0.1891	0.3819
ar-ur/ar	devtest	tfidf	0.2359	0.1987	0.4438
ar-ur/ur	dev	tfidf	0.2869	0.2647	0.4473
ar-ur/ur	devtest	tfidf	0.3087	0.2788	0.5154

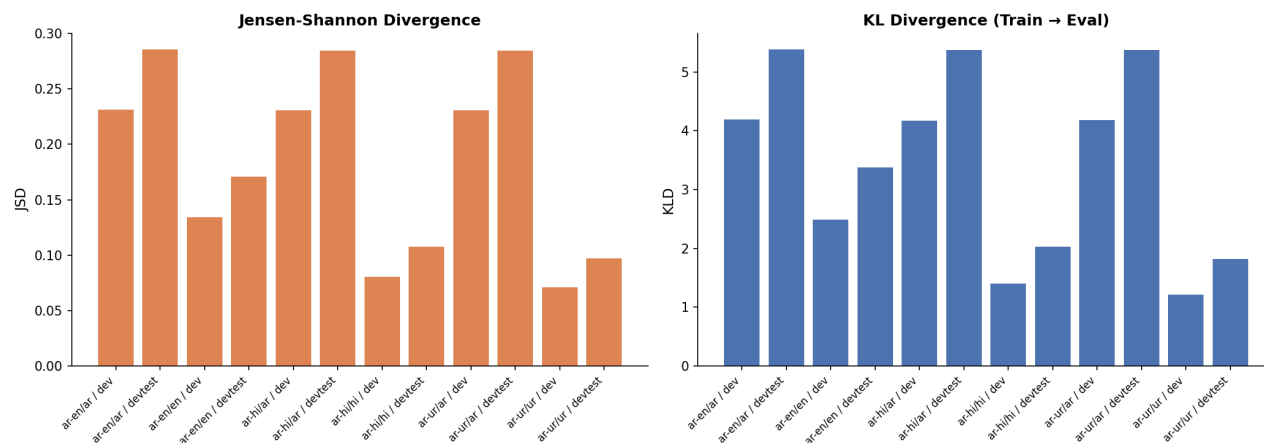


TF-IDF Semantic Similarity: Train vs Eval

2.9 Distribution Divergence (JSD / KLD)

direction	eval_split	jsd	kld
ar-en/ar	dev	0.231437	4.186342
ar-en/ar	devtest	0.285738	5.384518
ar-en/en	dev	0.134074	2.492235
ar-en/en	devtest	0.171053	3.372547
ar-hi/ar	dev	0.230446	4.174824
ar-hi/ar	devtest	0.284634	5.37357
ar-hi/hi	dev	0.080344	1.399125
ar-hi/hi	devtest	0.107353	2.023879
ar-ur/ar	dev	0.230655	4.176032

ar-ur/ar	devtest	0.284754	5.373013
ar-ur/ur	dev	0.071155	1.217745
ar-ur/ur	devtest	0.096892	1.82321



Jensen-Shannon & KL Divergence by Direction

2.10 Cross-Language Summary

language	avg_token_coverage	avg_oov_pct	avg_duplicate_pct	avg_semantic_sim
ar	89.89	10.11	0.0	0.2283
en	95.89	4.11	0.05	0.264
hi	97.8	2.2	0.05	0.418
ur	98.15	1.85	0.05	0.2978

Part 3: Overall Assessment

Direction: ar-en/ar

****Train → dev****

- Token coverage: 89.20% - OOV rate:

→ Representative: Needs review → No leakag

****Train → devtest****

- Token coverage: 90.70% - OOV rate:

→ Representative: Yes → No leakag

Direction: ar-en/en

****Train → dev****

- Token coverage: 95.43% - OOV rate:

→ Representative: Yes → No leakag

****Train → devtest****

- Token coverage: 96.36% - OOV rate:

→ Representative: Yes → No leakag

Direction: ar-hi/ar

****Train → dev****

- Token coverage: 89.08% - OOV rate:

→ Representative: Needs review

→ No leakag

****Train → devtest****

- Token coverage: 90.60%

- OOV rate:

→ Representative: Yes

→ No leakag

Direction: ar-hi/hi

****Train → dev****

- Token coverage: 97.52%

- OOV rate:

→ Representative: Yes

→ No leakag

****Train → devtest****

- Token coverage: 98.07%

- OOV rate:

→ Representative: Yes

→ No leakag

Direction: ar-ur/ar

****Train → dev****

- Token coverage: 89.10%

- OOV rate:

→ Representative: Needs review

→ No leakag

****Train → devtest****

- Token coverage: 90.64%

- OOV rate:

→ Representative: Yes

→ No leakag

Direction: ar-ur/ur

****Train → dev****

- Token coverage: 97.86%

- OOV rate:

→ Representative: Yes

→ No leakag

****Train → devtest****

- Token coverage: 98.44%

- OOV rate:

→ Representative: Yes

→ No leakag